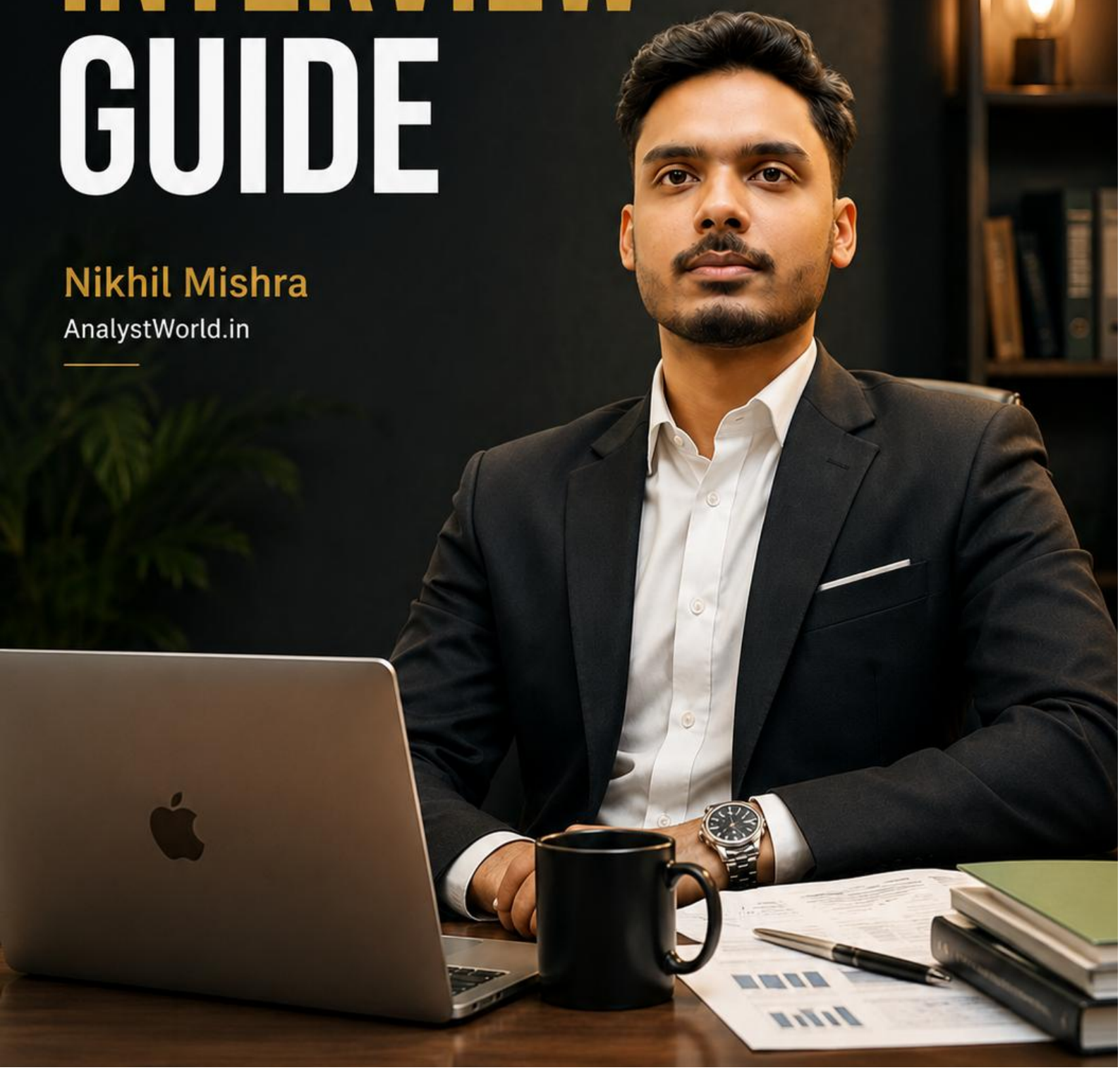

India's Only Book on Excel that can make you **Job ready**

EXCEL INTERVIEW GUIDE

Nikhil Mishra

AnalystWorld.in



Excel Interview Guide

...

A practical guide to thinking, solving, and communicating in Excel interviews.

Nikhil Mishra

A n a l y s t W o r l d . i n

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Author's Note

Over the last two years I have spent a large part of my time answering student questions about Excel. Some conversations happened in cohort classes with a few hundred people at once. Others were quiet one-on-one calls where a student was preparing for an interview scheduled the next morning. Across those two years, something kept surprising me.

Most students who came to me for interview help could actually use Excel. They had watched the videos, done the exercises, built the practice sheets. Ask them to write a VLOOKUP and they would write one. Ask them to build a Pivot Table and they would build one. But then I would change the question a little. I would say, tell me why you would use VLOOKUP here instead of INDEX-MATCH. Or, this lookup is returning #N/A even though the value looks right, what are three things you would check. Or, you receive a sales report every Monday with the same cleaning steps, what would you do differently.

At that point the tone of the conversation would often shift. The formula they had written thirty seconds ago suddenly did not feel connected to any real reasoning. They knew the syntax. They did not know the thinking behind it.

I saw this pattern often enough that I stopped treating it as a coincidence. It is not a lack of effort. Almost everyone I have mentored has worked hard, sometimes to a level I found humbling. The issue is that learning Excel functions and preparing for an Excel interview are two different activities, and most learning material only addresses the first one.

An interview is not a syntax test. An interviewer is trying to find out whether you can think about a business problem, decide which tool fits, use it correctly, notice when something is off, and explain your reasoning out loud to someone who may or may not be technical. Every one of those skills sits on top of the function knowledge, not underneath it. When we skip that layer during learning, the gap only shows up during an interview, when it is too late to close it.

That is the reason this book exists.

I did not write it as a function dictionary. There are enough of those. I wrote it as a book that walks a reader through what interviewers actually ask, why they ask it, and how to answer in a way that shows understanding. Every topic in this book starts with what you should know, then moves into interview questions with model answers, followed by a practical application and a

follow-up that a real interviewer would likely ask next. The intent is to help you build the layer that sits between knowing a function and using it well in front of a hiring manager.

I have tried to keep the writing honest. Where a concept is genuinely simple, I have not padded it. Where a concept is confusing, I have tried to explain the reasoning rather than the definition. The answers are written the way I would want a student to speak them out loud in an interview: clear, natural, without buzzwords, without over-selling. If an answer sounds too polished when you read it, please rephrase it in your own words before you rehearse it. An answer that sounds like yours is always stronger than one that sounds memorized.

A word about my own position. I am a Data Analyst by work and an educator by choice. Across the last two years I have had the privilege of working with more than twenty thousand students through cohorts and mentorship, and of guiding more than hundreds of them through their first Data Analyst placement. That number matters to me not because it is a milestone but because it is where most of the ideas in this book came from. Almost every question and answer here has been shaped by an actual conversation with an actual student who was preparing for an actual interview. If a section in this book feels unusually specific, it is probably because a version of that question came up more times than I can count.

This book is not a shortcut. It will not turn a beginner into an analyst in a weekend. What it can do is help you think, solve, and communicate the way an interviewer wants to see. That is the real work an interview is testing. Do that work well, and the rest tends to follow.

Nikhil Mishra
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Why This Book Exists

Knowing Excel functions is not the same as knowing Excel.

A function is a tool. Excel work is the thinking behind choosing the right tool, using it on messy real data, checking that the answer is sensible, and explaining what you did to someone who did not do it. Interviewers evaluate all of that, not just the syntax.

This book is built around that distinction. Every topic asks the same underlying questions. What problem does this solve? When would you use it, and when would you not? What can go wrong? How would you explain your choice in one clean sentence?

A good analyst does not memorize functions. A good analyst understands data, applies logic, respects business context, chooses the right tool, checks accuracy, and communicates clearly. If a book on Excel interview preparation ignores any of those, it is preparing you for the wrong test.

Use this book as a working companion, not a reference. Read a concept, then try the interview question aloud before checking the answer. Practice each formula on real data of your own. When a section feels obvious, do not skip it, because interviewers often build tricky questions on top of the basics.

The goal is simple. When you walk into your next interview, you should not be recalling. You should be thinking.

How to Use This Guide

This guide is a working handbook for Data Analyst students preparing for Excel interviews. It is not a beginner tutorial and it is not a definition dictionary. It mirrors the way interviewers actually ask questions.

The most effective way to use the guide is the following.

1. Read the short concept explanation at the start of each topic.
2. Look at the interview question first and try to answer it out loud before reading the model answer.
3. Compare your answer with the model. Focus on the reasoning, not on memorizing exact words.
4. Try every practical example in Excel. Reading a formula is not the same as writing one.
5. For every function, understand the problem it solves rather than only the syntax.
6. Practice explaining each concept in your own words in thirty to sixty seconds. Interviews are verbal, not written.
7. Revisit the follow-up questions. Interviewers rarely stop after your first answer.
8. Use the practical section at the end to simulate a live interview. Time yourself.

Do not try to finish the guide in one sitting. Cover one or two topics per day, apply them on real data, and then move on.

Contents

FRONT MATTER

Author's Note	1
Why This Book Exists	3
How to Use This Guide	4
Contents.....	5
The Interview Map.....	6
Opening Questions	8

SECTION ONE · THEORY & INTERVIEW QUESTIONS

01 Excel Fundamentals	12
02 Cell Referencing.....	15
03 Formatting and Number Formats	17
04 Excel Tables and Structured References	19
05 Sorting and Filtering.....	21
06 Data Validation.....	23
07 Conditional Formatting	25
08 Find and Replace, Go To Special, Text to Columns, Flash Fill	27
09 Named Ranges and Formula Auditing.....	29
10 Data Import and Data Consistency	31
11 Aggregation and Conditional Aggregation.....	33
12 Mathematical Functions	37
13 Statistical Functions.....	39
14 Logical Functions	42
15 Lookup and Reference Functions.....	45
16 Date and Time Functions.....	50
17 Text Functions.....	53
18 Dynamic Array Functions.....	56
19 Pivot Tables, Slicers and Timelines	59
20 Power Query	63

21 Data Cleaning.....	67
22 Data Visualization and Charts.....	71
23 Dashboards.....	74
24 What-If and Scenario Analysis.....	77
25 Macros and VBA.....	79

SECTION TWO · PRACTICAL INTERVIEW QUESTIONS

26 Formula-Based Problems	83
27 Lookup Problems	86
28 Data Cleaning Problems	89
29 Date and Time Problems.....	92
30 Pivot Table Problems	95
31 Power Query Problems.....	98
32 Dashboard Problems	100

BACK MATTER

How to Answer Project Questions.....	102
Final Revision Checklist.....	104
Last-Minute Interview Tips.....	106
A Closing Note.....	107
About the Author	108

The Interview Map

The following outline shows how the topics in this book connect. Use it as a checklist while you prepare and again on the morning of an interview.

EXCEL INTERVIEW PREPARATION

Foundations

- Interface and workbook basics
- Cells, ranges, data types
- Cell referencing: relative, absolute, mixed
- Excel Tables and structured references

Data management

- Sorting and filtering
- Data validation
- Conditional formatting
- Named ranges and formula auditing
- Data import and consistency

Formulas and functions

- Aggregation (SUM, SUMIF, SUMIFS, COUNT, AVERAGE)
- Mathematical (ROUND, INT, MOD, CEILING, FLOOR)
- Statistical (MEDIAN, STDEV, RANK, PERCENTILE, SUBTOTAL, AGGREGATE)
- Logical (IF, IFS, SWITCH, AND, OR, IFERROR, IFNA)
- Lookup (VLOOKUP, INDEX-MATCH, XLOOKUP, OFFSET, INDIRECT)
- Date and time
- Text
- Dynamic arrays (FILTER, UNIQUE, SORT, SEQUENCE)

Data cleaning

- Text cleaning, duplicates, missing values, data types

Pivot Tables

- Basics, grouping, calculated fields, slicers, timelines, charts

Power Query

Import and transform, merge and append, pivot and unpivot, refresh

Visualization and dashboards

What-if and scenario analysis

Macros and VBA

Business applications and reporting

Opening Questions

Most Excel interviews begin with a few general questions before moving into technical topics. They are used to judge your communication, confidence and honesty. The sample answers below are written for a fresher or early-career Data Analyst. Adapt them to your own background instead of memorizing them.

QUESTION 01

Tell me about yourself.

ANSWER

Keep this answer to about sixty to ninety seconds. Cover four points: your background, your interest in analytics, the skills you have built, and what you are looking for.

SAMPLE ANSWER

My name is Rahul. I am a commerce graduate, and I got interested in data during my final year while working on a small sales report project in Excel. Since then I have been learning analytics tools and I am now comfortable with Excel, Power Query, Pivot Tables and basic SQL. I have built a few practice projects around sales and HR data where I did the cleaning, analysis and reporting end to end. I am looking for an entry-level Data Analyst role where I can apply these skills on real business problems and continue learning.

QUESTION 02

Help me understand your profile.

ANSWER

This is a variant of "tell me about yourself" but the interviewer wants specifics: your education, tools, and any hands-on work. Do not repeat your resume line by line. Highlight two or three concrete things.

SAMPLE ANSWER

I have a background in commerce and started working with Excel during college for basic reporting. Over the last few months I have focused on data analysis: I use Excel for cleaning and reporting, Power Query for repeatable transformations, and Pivot Tables for summarization. I have built two practice projects, one on retail sales and one on HR attrition, where I cleaned the data, built KPIs, and created a simple dashboard. I try to communicate findings in business language, which is something I focus on because analysis alone does not help if the audience cannot understand it.

QUESTION 03**What projects have you worked on?****ANSWER**

Pick one project you know well. Explain the business problem, the data, what you did, and the outcome. Interviewers value depth over quantity.

SAMPLE ANSWER

One project I worked on was a retail sales analysis. I had around fifty thousand transaction records with product, region, date and revenue columns. The raw data had duplicate rows, inconsistent product names and a few missing dates. I cleaned it using Power Query and then built a Pivot Table dashboard showing revenue by region, top ten products and monthly trends. The main insight was that two regions were driving almost sixty percent of the revenue while a third region was underperforming on premium products. I presented this in a one-page Excel dashboard with slicers so the user could filter by category.

QUESTION 04**How confident are you with Excel?****ANSWER**

Be honest. Overselling is a common mistake and interviewers will test claims immediately with a practical question.

SAMPLE ANSWER

I would rate myself around seven or eight out of ten. I am very comfortable with formulas, Pivot Tables, lookups, Power Query and building basic dashboards. I have less experience with heavy VBA automation, so I would rate that lower. I am confident enough to solve most day-to-day analyst tasks and I know how to look up what I do not remember.

QUESTION 05**What are your strongest areas in Excel?****ANSWER**

Name two or three specific areas and briefly explain how you use them. Avoid listing everything, since that sounds rehearsed.

SAMPLE ANSWER

My strongest areas are lookup and reporting. I use XLOOKUP and INDEX-MATCH regularly to combine data from multiple sheets, and I rely on Pivot Tables and slicers to build interactive reports. I am also comfortable with Power Query for cleaning recurring reports, which has saved a lot of manual work in my practice projects.

QUESTION 06**Why do you use Excel when tools such as SQL or Power BI are available?****ANSWER**

This is a decision-making question. The interviewer wants to see if you understand where each tool fits.

SAMPLE ANSWER

Each tool has its place. SQL is best when the data lives in a database and I need to pull specific slices. Power BI is best for large interactive dashboards used by many people. Excel is the fastest choice for quick analysis, ad-hoc calculations, small to mid-sized datasets, and reports that a business user will edit or filter on their own. In real work I use them together: SQL to pull the data, Excel to explore and analyze it, and Power BI when the report needs to be shared widely.

SECTION 01

...

Theory & Interview Questions

Understand the concept. Explain the reasoning. Handle the follow-up.

01

Excel Fundamentals

WHAT YOU SHOULD KNOW

Excel is a spreadsheet application used to store, calculate, analyze and present data. A workbook is the file. Each workbook contains one or more worksheets. A worksheet is made of rows (numbered) and columns (lettered). The intersection of a row and a column is a cell, and a group of cells is a range. Excel supports different data types (text, numbers, dates, booleans, errors), and understanding these types is the foundation of every formula you will ever write.

INTERVIEW QUESTIONS

QUESTION 01

What is the difference between a workbook and a worksheet?

ANSWER

A workbook is the Excel file, saved with an extension such as .xlsx. A worksheet is a single tab inside the workbook where the data actually sits. One workbook can contain many worksheets, which is how analysts separate raw data, calculations and reports inside the same file.

EXAMPLE :

A monthly sales file might have three worksheets in one workbook: RawData, Calculations, and Dashboard.

QUESTION 02

What data types does Excel support and why does that matter?

ANSWER

Excel mainly works with text, numbers, dates and times, booleans (TRUE/FALSE), and error values (#N/A, #VALUE!, #REF!, #DIV/0!, #NAME?, #NULL!). Dates and times are stored internally as numbers, which is why you can add or subtract them.

It matters because most formula problems come from wrong data types. A number stored as text will not sum correctly. A date stored as text will not sort correctly. Cleaning data types is often the first job of an analyst.

QUESTION 03

How can you tell if a number is actually stored as text?

ANSWER

There are a few quick checks:

- The number appears left-aligned by default, while real numbers are right-aligned.
- A small green triangle usually appears in the top left of the cell.
- SUM or other math functions ignore the cell, giving a lower total than expected.
- ISNUMBER(A1) returns FALSE.

To fix it you can use Text to Columns, multiply by 1, or use the VALUE function.

QUESTION 04

What is the difference between a cell and a range?

ANSWER

A cell is a single box identified by a column letter and a row number, such as B4. A range is a group of cells, such as B4:B10 (a column range), B4:D4 (a row range), or B4:D10 (a rectangular block). Most functions accept a range as input.

QUESTION 05

What are Freeze Panes and when would you use them?

ANSWER

Freeze Panes keeps selected rows or columns visible while you scroll through the rest of the sheet. It is used mostly to keep headers visible while scrolling through long datasets.

USED WHEN :

You are reviewing 20,000 rows of sales data and you want the header row to stay visible so you always know what each column means.

QUESTION 06

What is the difference between Save and Save As?

ANSWER

Save writes changes to the existing file. Save As creates a new copy at a new location or with a new name, and does not affect the original. Save As is used to keep versions, change formats (for example xlsx to csv), or share a copy without exposing internal working sheets.

QUESTION 07

What are some Excel shortcuts an analyst should know?

ANSWER

A short set that covers most day-to-day analyst work:

- Ctrl + Arrow keys: jump to the last used cell in a direction.
- Ctrl + Shift + Arrow: select from the current cell to the last used cell.
- Ctrl + T: convert a range to an Excel Table.
- Ctrl + Shift + L: toggle filters on and off.
- Alt + =: insert a SUM automatically.
- F4: toggle absolute/relative reference while editing a formula.
- Ctrl + ;: insert today's date. Ctrl + Shift + ;: insert current time.
- Alt + Enter: line break inside a cell.

PRACTICAL APPLICATION

Excel fundamentals show up in every analyst task: laying out data with clean headers, freezing panes on long reports, checking that data types are correct before applying formulas, and using shortcuts to move through data quickly.

INTERVIEWER FOLLOW-UP

Q. What is the maximum number of rows and columns in an Excel worksheet?

A. A single worksheet supports 1,048,576 rows and 16,384 columns (XFD). This limit matters because if a dataset is close to that size, Excel becomes slow and Power Query or a database is usually a better choice.

02

Cell Referencing

WHAT YOU SHOULD KNOW

A cell reference lets a formula point to another cell rather than hard-coding a number. Excel supports three types of references: relative (A1), absolute (\$A\$1) and mixed (\$A1 or A\$1). Understanding when each type is used is the single most important formula skill. Almost every formula error in an analysts life traces back to a wrong reference type.

INTERVIEW QUESTIONS

QUESTION 01

What is cell referencing and why is it used?

ANSWER

Cell referencing means writing a formula that points to another cell instead of a fixed value. This lets the formula update automatically when the source cell changes and lets the same formula be copied down or across a range.

EXAMPLE:

If A1 contains 1000 and B1 contains 10%, the formula `=A1*B1` gives the discount amount. If A1 later changes to 2000, the formula recalculates automatically. Without references you would need to rewrite the formula every time.

QUESTION 02

What is the difference between relative and absolute referencing?

ANSWER

A relative reference (A1) shifts when the formula is copied. An absolute reference (\$A\$1) stays fixed no matter where the formula is copied.

EXAMPLE:

`=A2*B2` copied down becomes `=A3*B3`, `=A4*B4` ...

`=A2*$B$1` copied down becomes `=A3*$B$1`, `=A4*$B$1` ...

Use absolute when the formula needs to always pull from one fixed cell, such as a tax rate or a target.

QUESTION 03

What is a mixed reference?

ANSWER

A mixed reference locks only the row or only the column. \$A2 locks the column A but lets the row change. A\$2 locks the row 2 but lets the column change.

USED WHEN:

You are building a multiplication table or a two-way lookup grid where the formula must move down one axis and across the other.

QUESTION 04

How do you switch between reference types quickly?

ANSWER

Select the reference in the formula bar and press F4. Each press cycles through A1, \$A\$1, A\$1, \$A1, and back to A1. This is much faster than typing the dollar signs manually.

QUESTION 05

If you copy the formula =\$A2 from B2 to D5, what does it become?

ANSWER

The column is locked at A, but the row is relative. Copying from B2 to D5 moves the formula three rows down. The reference becomes =\$A5.

QUESTION 06

What is a 3D reference?

ANSWER

A 3D reference points to the same cell or range across multiple worksheets. It is written as SheetStart:SheetEnd!Range.

EXAMPLE:

`=SUM(Jan:Dec!B5)` adds cell B5 across every sheet from Jan to Dec.

It is useful when the same report layout exists on many monthly sheets.

PRACTICAL APPLICATION

- Applying a single tax rate stored in one cell across a long price list.
- Building a commission table where the rate cell is fixed but the sales cells change.
- Creating a two-variable data table where one input runs down and the other runs across.

INTERVIEWER FOLLOW-UP

Q. If you copy `=B2*C2` from row 2 to row 10 without any dollar signs, what happens?

A. Both references adjust relatively, so the formula in row 10 becomes $=B10*C10$. This is usually what you want when calculating a per-row value like line total = price times quantity.

03

Formatting and Number Formats

WHAT YOU SHOULD KNOW

Formatting changes how a value is displayed, not the underlying value. A cell may show 1,250.00 or \$1,250 or 1.25K while the stored number is still 1250. Understanding this difference prevents a very common mistake: assuming that a formatted number is the actual number.

INTERVIEW QUESTIONS

QUESTION 01

What is the difference between formatting a cell and changing its value?

ANSWER

Formatting changes only the appearance. If A1 contains 1250 and you format it as currency, it shows \$1,250 but the value is still 1250 for calculations. Changing the value replaces the number itself. This distinction matters because formatting cannot round or truncate for calculations; it only changes what the user sees.

QUESTION 02

What is the difference between the Number, Currency and Accounting formats?

ANSWER

All three display numbers, but they differ in alignment and symbol placement:

- Number: shows numbers with optional decimal places and comma separators, no currency symbol.
- Currency: places the currency symbol next to the number and can show negatives in red or brackets.
- Accounting: aligns the currency symbol at the left edge of the cell and aligns decimals in a column so the values line up cleanly in a financial report.

QUESTION 03

How do you create a custom number format?

ANSWER

Right-click, choose Format Cells, go to Custom, and enter a format code. The code uses four sections separated by semicolons: positive; negative; zero; text.

EXAMPLE :

```
| #,##0;[Red](#,##0);"-";@
```

This shows positive numbers with commas, negative numbers in red brackets, zero as a dash, and text as it is entered.

QUESTION 04

How can you display large numbers in thousands or millions without changing the value?

ANSWER

Use a custom format. To show a number in thousands with a K suffix, use:

```
| #,##0,"K"
```

Each comma after the last placeholder divides the display by 1,000. Two commas mean millions. The stored value stays the same.

QUESTION 05

Why is it a bad idea to merge cells in an analytical report?

ANSWER

Merged cells break most Excel features: sorting, filtering, Pivot Tables, structured references, and copy-paste. They are fine for titles or headers in a print report, but not for the data area. A cleaner alternative is Format Cells > Alignment > Center Across Selection, which gives the same visual look without merging.

QUESTION 06

What is conditional formatting and how does it differ from regular formatting?

ANSWER

Regular formatting is fixed. Conditional formatting changes automatically based on the cells value or a formula. It is used to highlight patterns, outliers, top values, duplicates, or missed targets.

EXAMPLE :

Highlighting all cells in a Sales column that fall below the target, so the manager immediately sees which reps are behind.

PRACTICAL APPLICATION

- Standardizing all financial reports with an Accounting format so numbers line up.
- Showing large revenue numbers in a KPI card as \$12.5M instead of 12,485,321.
- Using conditional formatting to highlight negative variances in a variance report.

04

Excel Tables and Structured References

WHAT YOU SHOULD KNOW

An Excel Table (Ctrl + T) is not just a formatted range. It is a defined object that expands automatically, has structured references, keeps formulas consistent, and works cleanly with Pivot Tables and Power Query. Tables are one of the most under-used Excel features among beginners.

INTERVIEW QUESTIONS

QUESTION 01

What is an Excel Table and why should an analyst use one?

ANSWER

An Excel Table is a named data range that behaves as one object. When new rows or columns are added, formulas, formatting, filters and connected Pivot Tables update automatically. It also enables structured references, which are easier to read than plain cell addresses.

USED WHEN :

You receive a report every month that grows in size, and you want the analysis to update without editing formula ranges every time.

QUESTION 02

What is a structured reference?

ANSWER

A structured reference uses table and column names instead of cell addresses. Instead of =SUM(B2:B100), you can write:

```
| =SUM(Sales[Revenue])
```

This is easier to read, does not break when rows are added, and makes the formula self-explanatory.

QUESTION 03

How is an Excel Table different from a normal range?

ANSWER

The main differences:

- Tables auto-expand when new rows or columns are added.
- Formulas inside a table auto-fill down all rows.
- Column headers stay visible while you scroll.
- Tables use structured references instead of cell addresses.
- Pivot Tables and Power Query linked to a table refresh with the new data automatically.

QUESTION 04

What is the difference between formatting a range as a table and just applying colors?

ANSWER

Applying colors is only visual. Converting to a table makes it a real data object with all the behaviors described above. If a Pivot Table is pointed at a colored range, it will not pick up new rows automatically. If it is pointed at a table, it will.

QUESTION 05

Are there any drawbacks to Excel Tables?

ANSWER

A few real ones:

- You cannot use array formulas that span multiple rows inside a table column.
- Tables do not support merged cells.
- Copy-pasting parts of a table can produce unexpected results because the table tries to maintain its structure.
- Very large tables (hundreds of thousands of rows) can slow down the workbook.

For most analyst work these drawbacks are minor compared to the benefits.

PRACTICAL APPLICATION

- Building a recurring monthly report where the data grows every month.
- Feeding a Pivot Table or Power Query with a source that auto-expands.
- Writing readable formulas like `=SUM(Orders[Amount])` instead of `=SUM(D2:D5000)`.

INTERVIEWER FOLLOW-UP

Q. If a Pivot Table is built on a plain range and new rows are added, will it show them after refresh?

A. No. It will only include the original range. Either the source range must be updated manually, or the source should be converted to an Excel Table before creating the Pivot Table.

05

Sorting and Filtering

WHAT YOU SHOULD KNOW

Sorting arranges rows in a chosen order. Filtering hides rows that do not match a condition, leaving the data underneath untouched. Both are fundamental to any analyst workflow. Advanced Filter goes further and can extract unique values or copy filtered rows to another location.

INTERVIEW QUESTIONS

QUESTION 01

What is the difference between sorting and filtering?

ANSWER

Sorting rearranges the order of rows based on values in one or more columns. Filtering hides rows that do not meet a condition, without changing the data. Sorting is destructive to the original order if you save the file; filtering is not, because a filter can be removed to bring everything back.

QUESTION 02

How do you perform a multi-level sort?

ANSWER

Go to Data > Sort. Add levels one by one. Each level defines a column and a direction. Excel applies the first level, then within each group applies the second, and so on.

EXAMPLE:

Sort by Region ascending, then by Revenue descending. Rows are grouped by region, and within each region the highest revenue appears first.

QUESTION 03

What is a custom sort?

ANSWER

A custom sort orders values in a user-defined sequence instead of alphabetical or numerical order. It is done by creating a Custom List (File > Options > Advanced > Edit Custom Lists) and then choosing that list in the Sort dialog.

USED WHEN :

Sorting months in calendar order (Jan, Feb, Mar) or priority values as High, Medium, Low, instead of alphabetical order.

QUESTION 04

What is the difference between a normal filter and an Advanced Filter?

ANSWER

A normal filter hides rows in place. Advanced Filter can copy filtered results to a new location, extract unique values, and use complex criteria written in a separate range including formulas.

USED WHEN :

You want a clean list of unique customer names extracted to another sheet without touching the original data.

QUESTION 05

What happens if you sort only one column of a wide dataset?

ANSWER

If Excel does not detect a table structure, it will sort only that column while leaving the others untouched. This scrambles the data because rows lose their alignment. Always select the whole range, or convert the data to an Excel Table, before sorting.

QUESTION 06

How would you extract only unique customer names from a large dataset?

ANSWER

Three common methods:

- Use the UNIQUE function if the version of Excel supports it: =UNIQUE(A2:A1000).
- Use Remove Duplicates from the Data tab.
- Use Advanced Filter with the Unique records only option.

Power Query is another option and is best when this must be done repeatedly on new data.

PRACTICAL APPLICATION

- Sorting a sales dataset by region then by revenue to identify top performers per region.
- Filtering a transactions table to show only refunds above a threshold.
- Using Advanced Filter to export a distinct list of active customers to a new sheet.

06

Data Validation

WHAT YOU SHOULD KNOW

Data Validation controls what a user can type into a cell. It is used to prevent bad entries, to standardize categories, and to build simple drop-down lists. It sits under Data > Data Validation.

INTERVIEW QUESTIONS

QUESTION 01

What is Data Validation and why is it important?

ANSWER

Data Validation is a rule attached to a cell that restricts what values can be entered. It is important because dirty input data leads to broken formulas, wrong Pivot Table results and inconsistent reports. Preventing bad data at entry is easier than cleaning it later.

QUESTION 02

How do you create a drop-down list in Excel?

ANSWER

Select the cell, go to Data > Data Validation, choose Allow: List, and either type the values separated by commas or point to a range that holds the list.

EXAMPLE:

For a Status column with only three valid values, use a list source of "Active,Inactive,Pending" so users cannot type variations like "active" or "act".

QUESTION 03

How can you make a drop-down list dynamic so it grows automatically?

ANSWER

Store the list of values inside an Excel Table. When new values are added to the table, the drop-down updates. Refer to the table column using a named range, since Data Validation does not accept structured references directly in some versions.

QUESTION 04

What are Input Message and Error Alert in Data Validation?

ANSWER

Input Message shows a small pop-up when the cell is selected, telling the user what is allowed. Error Alert appears when the user tries to enter something invalid. Error Alert has three styles: Stop (blocks entry), Warning (asks for confirmation), and Information (only informs).

QUESTION 05

How do you validate that a number is between a minimum and a maximum?

ANSWER

In Data Validation, choose Allow: Whole Number or Decimal, then set the Data option to "between" and provide the minimum and maximum. Any value outside the range is rejected based on the Error Alert setting.

USED WHEN :

Restricting a Discount % column so users can only enter values between 0 and 30.

QUESTION 06

Can Data Validation prevent duplicates?

ANSWER

Yes, using a custom formula. For column A, use:

```
=COUNTIF($A$2:$A$1000,A2)=1
```

This blocks entry of any value already present in the column. It is commonly used for Employee ID or Invoice Number fields.

PRACTICAL APPLICATION

- Standardizing region names using a drop-down so reporting stays consistent.
- Preventing incorrect date ranges in a leave application form.
- Blocking duplicate customer IDs in a master data sheet.

07

Conditional Formatting

WHAT YOU SHOULD KNOW

Conditional Formatting changes the appearance of a cell based on its value or a formula. It is a visual analysis tool: it does not change the data, only how it looks. Types include highlight rules, top and bottom rules, data bars, color scales, icon sets, and formula-based rules.

INTERVIEW QUESTIONS

QUESTION 01

What is Conditional Formatting used for in a Data Analyst role?

ANSWER

It is used to draw attention to patterns and exceptions without needing extra calculations: highlighting missed targets, flagging duplicates, showing top and bottom values, and comparing performance across a range.

QUESTION 02

What is the difference between a rule based on cell value and a rule based on a formula?

ANSWER

A cell value rule checks the value of the cell itself, for example "greater than 1000". A formula rule can use any logic including references to other cells, allowing more complex conditions such as highlighting an entire row when one column meets a condition.

EXAMPLE:

| =\$D2>\$E2

Applied to the whole row range, this highlights every row where actual sales exceed target sales.

QUESTION 03

How would you highlight the entire row of a dataset based on a condition?

ANSWER

Select the whole data range (excluding headers). Go to Conditional Formatting > New Rule > Use a formula. Write a formula that starts with a dollar sign to lock the column, such as `=D2="Overdue"`. Apply the format and click OK. The dollar sign is essential; without it, only the matching cell will highlight, not the whole row.

QUESTION 04

How would you highlight duplicates in a list?

ANSWER

Select the range, go to Conditional Formatting > Highlight Cells Rules > Duplicate Values. This highlights every value that appears more than once. For a more controlled version, use a formula:

```
=COUNTIF($A$2:$A$1000,A2)>1
```

QUESTION 05

What is the difference between Data Bars, Color Scales, and Icon Sets?

ANSWER

Data Bars draw a horizontal bar inside each cell, longer bars for larger values, useful for quick magnitude comparison. Color Scales apply a color gradient across the range, showing low to high through color. Icon Sets place small icons (arrows, traffic lights) based on value thresholds. They are all built-in visual patterns and do not require formulas.

QUESTION 06

How do you remove Conditional Formatting from a sheet?

ANSWER

Go to Conditional Formatting > Clear Rules > Clear Rules from Selected Cells or Clear Rules from Entire Sheet. To edit or delete a specific rule, use Manage Rules from the same menu.

PRACTICAL APPLICATION

- Highlighting sales reps who missed their target in red.
- Flagging overdue invoices based on due date compared with today.
- Using color scales on a variance column to see high and low variance regions quickly.

INTERVIEWER FOLLOW-UP

Q. If two conditional formatting rules apply to the same cell, which wins?

A. The rule that is higher in the Manage Rules list is applied first. You can drag rules to change priority, and you can check "Stop If True" to prevent later rules from applying.

08

Find and Replace, Go To Special, Text to Columns, Flash Fill

WHAT YOU SHOULD KNOW

These are small tools individually, but together they cover a large part of everyday data cleaning. Every interview panel expects an analyst to know when to reach for each of them.

INTERVIEW QUESTIONS

QUESTION 01

What is the difference between Find and Find & Replace?

ANSWER

Find (Ctrl + F) locates values in the sheet. Find & Replace (Ctrl + H) locates values and replaces them with something else. Both support wildcards, match case and match entire cell contents options.

QUESTION 02

What are wildcards in Find and Replace?

ANSWER

Wildcards let you match patterns instead of exact text:

- * matches any number of characters.
- ? matches exactly one character.
- ~ escapes a wildcard, so ~* means the actual asterisk character.

EXAMPLE:

Searching for M* will match Mumbai, Mangalore and Madurai. Searching for M??? will match only four-letter words starting with M.

QUESTION 03

What is Go To Special and when is it useful?

ANSWER

Go To Special (Ctrl + G, then Special) selects cells based on their content: blanks, constants, formulas, errors, visible cells only, and more. It is used to quickly find or act on cells that share a property.

USED WHEN :

You have a large dataset with random blank cells and you want to fill them all with zero. Select the range, Go To Special > Blanks, type 0, then press Ctrl + Enter to fill all selected blanks at once.

QUESTION 04

What is Text to Columns and what are its common uses?

ANSWER

Text to Columns splits the contents of one column into several columns using a delimiter (comma, tab, space, or custom) or a fixed width. It can also convert text-formatted numbers or dates into their proper types by running through the wizard and choosing the correct data format.

EXAMPLE :

A Full Name column containing "Rahul Sharma" can be split into First Name and Last Name using space as the delimiter.

QUESTION 05

What is Flash Fill and how is it different from Text to Columns?

ANSWER

Flash Fill (Ctrl + E) fills a column by recognizing a pattern from your first few examples. It is faster than Text to Columns for simple extractions but works only when the pattern is clear.

EXAMPLE :

If A2 contains "rahul.sharma@company.com" and you type "rahul.sharma" in B2, Flash Fill can populate B3, B4 and so on by detecting the pattern. It is a shortcut for pattern-based text extraction that would otherwise need LEFT, RIGHT or MID formulas.

QUESTION 06

When would you prefer formulas over Flash Fill?

ANSWER

Formulas are better when the data will change or grow, when the transformation must be repeatable, or when the pattern is not obvious enough for Flash Fill to detect. Flash Fill is a one-time action; a formula updates automatically when the source changes.

PRACTICAL APPLICATION

- Replacing all "NA" text values with blanks before importing into a Pivot Table.
- Selecting all cells with errors using Go To Special so they can be reviewed together.
- Splitting a "Date - Region" concatenated column into two clean columns.

09

Named Ranges and Formula Auditing

WHAT YOU SHOULD KNOW

A named range gives a meaningful label to a cell or range so formulas become easier to read. Formula Auditing tools help you trace how a formula depends on other cells and find the source of errors in a complex workbook.

INTERVIEW QUESTIONS

QUESTION 01

What is a Named Range and why would you use one?

ANSWER

A Named Range is a friendly name assigned to a cell or range, defined through the Name Box or Formulas > Define Name. Instead of `=B2*Sheet1!$G$1`, you can write `=B2*TaxRate`. It makes formulas more readable, less error-prone, and easier to hand off to another analyst.

QUESTION 02

What is a dynamic named range?

ANSWER

A dynamic named range automatically adjusts its size when data is added or removed. Traditionally it was built using `OFFSET` or `INDEX`. In modern Excel, converting the source to an Excel Table achieves the same effect more cleanly, because a table column reference is already dynamic.

EXAMPLE (USING INDEX):

```
=Sheet1!$A$2:INDEX(Sheet1!$A:$A,COUNTA(Sheet1!$A:$A))
```

QUESTION 03

What is the scope of a Named Range?

ANSWER

Scope defines where the name can be used. It can be workbook-level (usable in any sheet of the workbook) or worksheet-level (usable only in the sheet where it was defined). Worksheet-level

names are useful when different sheets have similar names like SalesData that should not conflict.

QUESTION 04

What are Trace Precedents and Trace Dependents?

ANSWER

Trace Precedents draws arrows from a cell back to the cells whose values feed into its formula. Trace Dependents draws arrows to the cells whose formulas depend on the selected cell. They are used to debug complex workbooks where you cannot tell at a glance where a wrong number came from.

QUESTION 05

What does Evaluate Formula do?

ANSWER

Evaluate Formula (Formulas > Evaluate Formula) walks through a complex formula one step at a time, showing the intermediate result at each step. It is the fastest way to understand where a large nested formula goes wrong.

QUESTION 06

What is the Watch Window?

ANSWER

The Watch Window (Formulas > Watch Window) pins selected cells so you can monitor their values while working elsewhere in the workbook. It is used when key totals or KPIs live on a different sheet and you want to see them change as you edit the source data.

PRACTICAL APPLICATION

- Using =Discount instead of =\$B\$1 in a large pricing model so the formula tells its own story.
- Using Trace Precedents to debug a KPI that is showing the wrong number.
- Using Evaluate Formula to walk through a five-level nested IF and find where the logic breaks.

10

Data Import and Data Consistency

WHAT YOU SHOULD KNOW

Data usually arrives from outside sources: CSV files, text files, databases, or web downloads. How you import it decides how much cleaning you will need afterwards. For repeatable imports, Power Query is the professional choice.

INTERVIEW QUESTIONS

QUESTION 01

How do you import a CSV file into Excel?

ANSWER

Three common ways:

- Open the CSV directly. Excel opens it as a worksheet, but any leading zeros or long numeric strings may be lost.
- Data > From Text/CSV. This uses Power Query and lets you set data types before loading.
- Copy paste and then use Text to Columns to split. This is quickest but manual.

For any repeatable process, use From Text/CSV so the steps can be refreshed later.

QUESTION 02

Why do leading zeros disappear when a CSV is opened in Excel?

ANSWER

Excel auto-detects columns as numbers and drops the leading zeros. To keep them, import through Power Query and set the column type to Text before loading, or open the file through Data > From Text/CSV and set the correct type in the import dialog.

QUESTION 03

What is a delimiter and which delimiters are common?

ANSWER

A delimiter is the character used to separate values in a text file. The most common ones are comma (CSV), tab (TSV), pipe (|), and semicolon. You must know the delimiter to import the file correctly, otherwise all data ends up in one column.

QUESTION 04

What checks would you run on newly imported data before analysis?

ANSWER

A short professional checklist:

- Row count matches the source.
- Column headers are clean and unique.
- Data types are correct (numbers as numbers, dates as dates).
- No duplicate primary keys.
- Missing values are identified and handled.
- Text values are trimmed and consistent in case.
- Categorical columns have a limited, expected set of values.

QUESTION 05

How do you refresh imported data?

ANSWER

For a Power Query connection, right-click the query and choose Refresh, or press Alt + F5. You can also set Refresh All (Ctrl + Alt + F5) to update every connection in the workbook. Automatic refresh on file open can be enabled in the query properties.

QUESTION 06

What is data consistency and why does it matter?

ANSWER

Data consistency means that the same real-world value is represented the same way everywhere. "Mumbai", "mumbai" and "MUMBAI" should either be treated as the same city or the difference is intentional. Inconsistent data causes Pivot Tables to split what should be one group into three, causes lookups to fail, and produces incorrect counts.

PRACTICAL APPLICATION

- Importing a monthly bank statement CSV into Excel through Power Query so leading zeros in account numbers survive.
- Cleaning inconsistent city names before building a regional sales report.

- Automating the monthly refresh of imported files by parameterizing the file path in Power Query.

11

Aggregation and Conditional Aggregation

WHAT YOU SHOULD KNOW

Aggregation means summarizing many values into one number: total, count, average, minimum, maximum. Conditional aggregation adds a filter: aggregate only where a condition is met. In Excel this is done through the SUM, COUNT and AVERAGE families and their conditional variants: SUMIF, SUMIFS, COUNTIF, COUNTIFS, AVERAGEIF, AVERAGEIFS. Interviewers almost always test this because it is at the heart of business reporting.

INTERVIEW QUESTIONS

QUESTION 01

What is the difference between SUM, SUMIF and SUMIFS?

ANSWER

SUM adds all numbers in a range. SUMIF adds numbers that meet a single condition. SUMIFS adds numbers that meet one or more conditions.

EXAMPLE:

<code>=SUM(D2:D100)</code>	total revenue
<code>=SUMIF(B2:B100,"West",D2:D100)</code>	revenue only for the West region
<code>=SUMIFS(D2:D100,B2:B100,"West",C2:C100,"Laptop")</code>	revenue for West and Laptop

SUMIFS is preferred in most business scenarios because real questions almost always have more than one filter.

QUESTION 02

What is the syntax difference between SUMIF and SUMIFS?

ANSWER

The argument order changes:

<code>SUMIF(range, criteria, sum_range)</code>
<code>SUMIFS(sum_range, criteria_range1, criteria1, criteria_range2, criteria2, ...)</code>

In SUMIFS the sum range comes first. This trips people up in interviews, so it is worth memorizing.

QUESTION 03

What is the difference between COUNT, COUNTA and COUNTBLANK?

ANSWER

COUNT counts only numeric values in the range. COUNTA counts any non-empty cell, including text and errors. COUNTBLANK counts empty cells.

EXAMPLE:

In a range that contains 100, "Yes", "", 50 and #N/A: COUNT returns 2, COUNTA returns 4, COUNTBLANK returns 1.

QUESTION 04

What is the difference between COUNTIF and COUNTIFS?

ANSWER

COUNTIF counts cells that meet one condition. COUNTIFS counts cells that meet multiple conditions across different ranges.

=COUNTIF(B2:B100, "West")	orders from West
=COUNTIFS(B2:B100, "West", D2:D100, ">1000")	orders from West above 1000

QUESTION 05

How do you use wildcards in SUMIFS or COUNTIFS?

ANSWER

Wildcards work in the criteria of these functions:

- * matches any number of characters.
- ? matches exactly one character.

EXAMPLE:

=COUNTIF(A2:A100, "*Ltd*")	counts all company names containing "Ltd"
----------------------------	---

QUESTION 06

How would you use comparison operators inside SUMIFS?

ANSWER

Comparison operators go inside quotes:

=SUMIFS(D2:D100, C2:C100, ">=" & E1, C2:C100, "<=" & E2)
--

This sums values in D where the value in C is between E1 and E2. The ampersand concatenates the operator with the cell reference.

QUESTION 07

What is the difference between AVERAGE, AVERAGEIF and AVERAGEIFS?

ANSWER

Same pattern as SUM. AVERAGE gives the mean of all numbers. AVERAGEIF gives the mean of values meeting one condition. AVERAGEIFS gives the mean under multiple conditions. AVERAGEIFS is the modern default.

QUESTION 08

What is the difference between MIN and MAX versus SMALL and LARGE?

ANSWER

MIN returns the smallest value. MAX returns the largest value. SMALL and LARGE return the k-th smallest or largest value, so you can ask for the second smallest or the third largest.

```
=LARGE(D2:D100,3)    the 3rd highest revenue
=SMALL(D2:D100,2)    the 2nd lowest revenue
```

Used when building "top 3" or "bottom 5" reports.

QUESTION 09

Why might SUM return the wrong total on a column that clearly contains numbers?

ANSWER

Common reasons:

- Some values are stored as text, so SUM ignores them.
- There are hidden rows containing values that are still included.
- The range does not cover all rows, especially after new data was added.
- There are error values in the range, which some functions propagate.

Convert text-numbers using VALUE or Text to Columns, and prefer Excel Tables so the range auto-expands.

QUESTION 10

When would you use SUMPRODUCT instead of SUMIFS?

ANSWER

SUMPRODUCT multiplies corresponding elements of ranges and sums the result. It is more flexible than SUMIFS because it can handle array logic and complex conditions in one formula.

```
=SUMPRODUCT((B2:B100="West")*(D2:D100>1000)*D2:D100)
```

For simple totals SUMIFS is faster and easier to read. Use SUMPRODUCT when the logic cannot be expressed cleanly with SUMIFS, for example when you need to combine columns arithmetically inside the condition.

PRACTICAL APPLICATION

- Calculating total revenue by region and product using SUMIFS.
- Counting how many customers placed more than five orders using COUNTIFS.
- Computing the average deal size for a segment using AVERAGEIFS.
- Finding the top three revenue values using LARGE for a leaderboard.

INTERVIEWER FOLLOW-UP

Q. What is the difference between the criteria "5" and 5 in SUMIF?

A. In SUMIF both usually work because the function coerces the criteria. But when the criteria uses an operator such as ">5" it must be a string. Passing >5 without quotes is a syntax error.

12

Mathematical Functions

WHAT YOU SHOULD KNOW

These are supporting functions used constantly in financial and operational calculations. Analysts should know when each is appropriate because the choice can change reported numbers.

INTERVIEW QUESTIONS

QUESTION 01

What is the difference between ROUND, ROUNDUP and ROUNDDOWN?

ANSWER

ROUND rounds to the nearest value using standard rules. ROUNDUP always rounds away from zero. ROUNDDOWN always rounds toward zero, regardless of decimal value.

=ROUND(4.4,0)	returns 4
=ROUND(4.6,0)	returns 5
=ROUNDUP(4.1,0)	returns 5
=ROUNDDOWN(4.9,0)	returns 4

ROUNDUP is used for pricing and billing where you cannot under-charge. ROUNDDOWN is used for capacity or slot calculations where you cannot exceed the limit.

QUESTION 02

What is the difference between INT and TRUNC?

ANSWER

Both return the integer part of a number, but they differ for negatives. INT rounds toward negative infinity, so $\text{INT}(-2.3) = -3$. TRUNC just drops the decimal, so $\text{TRUNC}(-2.3) = -2$. For positive numbers they behave the same.

QUESTION 03

What is the difference between CEILING and FLOOR?

ANSWER

CEILING rounds a number up to the nearest multiple of a given significance. FLOOR rounds down. They are used to snap numbers to a step size.

`=CEILING(23,5)` returns 25

`=FLOOR(23,5)` returns 20

Used when calculating billing slabs, freight brackets, or time intervals rounded to 15 minutes.

QUESTION 04

What does MOD do and when is it useful?

ANSWER

MOD returns the remainder after division.

`=MOD(10,3)` returns 1

It is commonly used to alternate rows, group items into buckets, or check divisibility. For example `=MOD(ROW(),2)` returns 0 for even rows and 1 for odd, useful in banded formatting.

QUESTION 05

What is ABS and when would you use it?

ANSWER

ABS returns the absolute value of a number, always positive. It is used when calculating variance size regardless of direction, or when the sign of a number should be ignored.

`=ABS(-125)` returns 125

QUESTION 06

How would you round a number to the nearest 50?

ANSWER

Use MROUND, or CEILING/FLOOR with a significance of 50.

`=MROUND(237,50)` returns 250

`=CEILING(237,50)` returns 250

`=FLOOR(237,50)` returns 200

PRACTICAL APPLICATION

- Rounding invoice amounts to two decimals for financial reports.
- Rounding call handling time up to the nearest 15 seconds for billing.
- Using MOD to identify every 4th row for sampling.

13

Statistical Functions

WHAT YOU SHOULD KNOW

Statistical functions summarize a distribution beyond the average. Interviewers ask about them because business data rarely follows a nice bell curve and a single average often hides the story.

INTERVIEW QUESTIONS

QUESTION 01

What is the difference between MEAN, MEDIAN and MODE?

ANSWER

MEAN (AVERAGE) is the arithmetic average. MEDIAN is the middle value when data is sorted. MODE returns the value that appears most often. The mean is sensitive to outliers; the median is not. If a company reports a mean salary of 12L but a median of 6L, most employees earn well below the mean.

QUESTION 02

What is the difference between STDEV.S and STDEV.P?

ANSWER

STDEV.S is for a sample of the data. STDEV.P is for the entire population. Use STDEV.S when the data represents a subset and you want to estimate the wider population variability. Use STDEV.P when the data is the full universe you care about. The math differs slightly: STDEV.S divides by $n-1$, STDEV.P by n .

QUESTION 03

What is the difference between VAR.S and VAR.P?

ANSWER

Same distinction as STDEV. VAR.S is for a sample. VAR.P is for a population. Variance is the square of standard deviation and is used in more advanced statistical analysis such as ANOVA.

QUESTION 04

What is the difference between RANK.EQ and RANK.AVG?

ANSWER

Both give the rank of a value in a list. If there are ties, RANK.EQ gives each tied value the same rank (the highest in the group) and skips the next ranks. RANK.AVG averages the ranks of the tied values.

EXAMPLE:

For values 100, 90, 90, 80: RANK.EQ ranks them 1, 2, 2, 4. RANK.AVG ranks them 1, 2.5, 2.5, 4.

QUESTION 05**What are QUARTILE and PERCENTILE?****ANSWER**

QUARTILE splits sorted data into four equal parts and returns the boundary of the requested quarter (0 = min, 1 = 25%, 2 = median, 3 = 75%, 4 = max). PERCENTILE gives the value below which a chosen percentage of the data falls.

```
| =PERCENTILE(D2:D100,0.9)    revenue below which 90% of orders fall
```

Used for customer segmentation, salary bands, and SLA measurements.

QUESTION 06**What is FREQUENCY used for?****ANSWER**

FREQUENCY counts how many values fall into given bins. It is used to build histograms or distribution tables. It is entered as an array formula in older Excel or spills automatically in newer versions.

QUESTION 07**What is the difference between SUBTOTAL and SUM?****ANSWER**

SUBTOTAL ignores hidden rows and other SUBTOTAL formulas, so it plays well with filtered data. SUM adds everything including hidden rows.

```
| =SUBTOTAL(9,D2:D100)    sums only visible rows after filtering
```

The first argument is a function code: 9 for SUM, 1 for AVERAGE, 3 for COUNTA, 2 for COUNT, and so on.

QUESTION 08**What is AGGREGATE and how is it different from SUBTOTAL?****ANSWER**

AGGREGATE is a newer function that supports more operations (including MEDIAN, LARGE, PERCENTILE) and can ignore errors and hidden rows. SUBTOTAL cannot ignore errors on its own. When a range may contain #N/A or #DIV/0, AGGREGATE is safer.

`=AGGREGATE(9,6,D2:D100)` sums D2:D100 ignoring errors and hidden rows

QUESTION 09

When would you use PERCENTILE for customer segmentation?

ANSWER

If you want to define your top 20% of customers by revenue, you can compute PERCENTILE(revenue, 0.8) and label anyone above that value as top-tier. This is more objective than picking a manual cutoff.

PRACTICAL APPLICATION

- Reporting median salary alongside mean salary in an HR analysis to show pay skew.
- Using STDEV.S on daily sales to measure demand stability.
- Using SUBTOTAL on filtered lists so the total updates as filters change.
- Using AGGREGATE to compute totals safely on a column that contains occasional errors.

INTERVIEWER FOLLOW-UP

Q. If most of your customers place small orders but a few place very large ones, which is more useful, mean or median?

A. Median. The mean is pulled up by the large orders and misrepresents the typical customer. Report both, so leadership sees the average value and the typical value.

14

Logical Functions

WHAT YOU SHOULD KNOW

Logical functions let you write business rules inside Excel: incentive slabs, grading bands, eligibility flags, approval logic. IF is the core function. IFS, SWITCH, AND, OR, NOT, IFERROR and IFNA extend it. Nested IF works but is hard to maintain; IFS or SWITCH is usually cleaner.

INTERVIEW QUESTIONS

QUESTION 01

What is the IF function and how is it structured?

ANSWER

IF checks a condition and returns one value if TRUE and another if FALSE.

```
| =IF(logical_test, value_if_true, value_if_false)
```

EXAMPLE:

```
| =IF(D2>10000, "High", "Low")
```

It is the building block of almost every business rule in Excel.

QUESTION 02

What is a nested IF and what are its problems?

ANSWER

A nested IF places one IF inside another to handle more than two possible outcomes. It works but becomes hard to read past three levels.

```
| =IF(D2>10000, "High", IF(D2>5000, "Medium", "Low"))
```

Problems: hard to debug, easy to mis-place a bracket, and hard to change later. Prefer IFS or a lookup table for many bands.

QUESTION 03

How is IFS different from nested IF?

ANSWER

IFS lets you list conditions and results in pairs, without nesting.

```
=IFS(D2>10000,"High",D2>5000,"Medium",TRUE,"Low")
```

It reads top to bottom and returns the first matching result. TRUE at the end works as a catch-all. It is easier to read and edit than nested IF.

QUESTION 04

When would you use SWITCH instead of IFS?

ANSWER

SWITCH is best when the condition compares one value against a list of exact matches. IFS is best when the conditions are inequalities.

```
=SWITCH(A2,"North",1,"South",2,"East",3,"West",4,0)
```

SWITCH is cleaner for lookups against a fixed list of codes.

QUESTION 05

What is the difference between AND, OR and NOT?

ANSWER

AND returns TRUE only when every condition is TRUE. OR returns TRUE when at least one condition is TRUE. NOT reverses TRUE and FALSE.

EXAMPLE:

```
=IF(AND(D2>1000, E2="Paid"), "Eligible", "Not Eligible")
```

These are combined inside IF to express compound business rules.

QUESTION 06

What is the difference between IFERROR and IFNA?

ANSWER

IFERROR catches every error type (#N/A, #VALUE!, #DIV/0!, #REF!, #NAME?, #NUM!, #NULL!). IFNA only catches #N/A. Use IFNA when you specifically want to handle a missing lookup value but still see other errors so you can fix real formula problems.

```
=IFERROR(VLOOKUP(...), "Not Found")    hides every error
```

```
=IFNA(VLOOKUP(...), "Not Found")        hides only #N/A
```

QUESTION 07

Why is it dangerous to wrap every formula in IFERROR?

ANSWER

IFERROR hides all errors, including ones that are caused by real formula mistakes. You may think the report is working while important calculations are silently producing "Not Found". Use IFERROR only when you have decided that a specific error type is acceptable and expected.

QUESTION 08**How do you build a grading system in Excel?****ANSWER**

Two clean approaches:

- Use IFS: =IFS(A2>=90,"A",A2>=75,"B",A2>=60,"C",TRUE,"D")
- Use a lookup table with VLOOKUP or XLOOKUP in approximate match mode. Store the lower bound of each grade in a table; the lookup finds the correct row and returns the grade.

The lookup approach is preferred when the bands may change, since editing the table is easier than editing a formula.

PRACTICAL APPLICATION

- Marking sales as target achieved or not achieved.
- Building an incentive calculator with slab-based rules.
- Wrapping VLOOKUP in IFNA so missing IDs show "New Customer" instead of #N/A.

INTERVIEWER FOLLOW-UP

Q. If IFS has no matching condition and no TRUE catch-all, what happens?

A. IFS returns #N/A. That is why analysts add TRUE as the final condition with a default value.

15

Lookup and Reference Functions

WHAT YOU SHOULD KNOW

Lookup functions pull a value from one range based on a match in another. VLOOKUP is the classic. INDEX-MATCH is more flexible. XLOOKUP is the modern replacement. HLOOKUP works on horizontal data. OFFSET, INDIRECT, CHOOSE, ROW and COLUMN are supporting functions used in advanced formulas. Almost every Data Analyst interview will have at least one lookup question.

INTERVIEW QUESTIONS

QUESTION 01

What problem does VLOOKUP solve?

ANSWER

VLOOKUP looks for a value in the first column of a table and returns a value from a chosen column in the same row. It is used to bring information from one dataset into another based on a shared key such as Employee ID or Product Code.

```
=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
```

EXAMPLE:

```
=VLOOKUP(A2, EmployeeMaster, 3, FALSE)
```

This finds the value in A2 (an Employee ID) inside the first column of the EmployeeMaster table and returns the value from the third column, using an exact match.

QUESTION 02

What is the difference between exact match and approximate match in VLOOKUP?

ANSWER

Exact match (FALSE or 0) returns a value only if the exact lookup value is found. Approximate match (TRUE or 1) finds the largest value less than or equal to the lookup value, assuming the first column is sorted ascending. Approximate match is used mainly for banded lookups like tax slabs and grading.

QUESTION 03

What are the main limitations of VLOOKUP?

ANSWER

Limitations that come up in interviews:

- The lookup column must be the first column of the table range.
- It cannot look to the left of the lookup column.
- The column index is a hard-coded number; inserting columns can break the formula.
- It only returns the first match; duplicates are ignored.
- The default TRUE range_lookup is dangerous because if the data is not sorted it returns wrong results silently.

These are the reasons INDEX-MATCH and XLOOKUP became popular.

QUESTION 04

What is INDEX-MATCH and why is it preferred over VLOOKUP?

ANSWER

INDEX returns a value from a specified position in a range. MATCH returns the position of a value in a range. Combined, they perform a lookup that can go in any direction.

```
=INDEX(return_column, MATCH(lookup_value, lookup_column, 0))
```

ADVANTAGES OVER VLOOKUP:

- The return column does not need to be to the right of the lookup column.
- Inserting columns does not break the formula.
- It can be faster on very large datasets.

QUESTION 05

What is XLOOKUP and how is it better than VLOOKUP?

ANSWER

XLOOKUP is a modern lookup function that combines the best of VLOOKUP and INDEX-MATCH.

```
=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found],  
[match_mode], [search_mode])
```

ADVANTAGES:

- Can look left, right, up or down.
- Built-in if_not_found argument replaces IFERROR wrapping.
- Supports approximate match without requiring sorted data.
- Can search from top or bottom to handle duplicates.

It is now the default lookup function for analysts using Microsoft 365 or Excel 2021.

QUESTION 06

What is the difference between INDEX-MATCH and XLOOKUP?

ANSWER

Both can do the same lookups. XLOOKUP is shorter, easier to read, and has a built-in not-found handler. INDEX-MATCH is available in all Excel versions, so it is still needed when sharing files with users on older Excel. Choose XLOOKUP for personal work, INDEX-MATCH for compatibility.

QUESTION 07

What is HLOOKUP and when is it used?

ANSWER

HLOOKUP is the horizontal version of VLOOKUP. It searches the first row of a range and returns a value from a specified row below. It is used when the data is laid out horizontally with headers across the top row. XLOOKUP handles both directions and is preferred in modern Excel.

QUESTION 08

How do you perform a two-way lookup?

ANSWER

A two-way lookup finds a value at the intersection of a row and a column, based on two criteria.

```
=INDEX(matrix, MATCH(rowKey, rowLabels, 0), MATCH(colKey, colLabels, 0))
```

Or with XLOOKUP:

```
=XLOOKUP(rowKey, rowLabels, XLOOKUP(colKey, colLabels, matrix))
```

USED WHEN:

Looking up the salary of a specific role at a specific experience level in a pay-grid table.

QUESTION 09

How would you handle a lookup with multiple criteria?

ANSWER

Options:

- Concatenate the criteria into one helper column and lookup on that key.
- Use INDEX with MATCH in array form: =INDEX(return, MATCH(1, (crit1=range1)*(crit2=range2), 0)) entered as an array formula.
- Use XLOOKUP with concatenation: =XLOOKUP(crit1 & crit2, range1 & range2, return).
- Use FILTER for multi-criteria retrieval when multiple results are needed.

QUESTION 10

Your XLOOKUP is returning #N/A even though the value appears to exist. What would you check?

ANSWER

Common causes of a phantom #N/A:

- Leading or trailing spaces in either the lookup value or the lookup array. Use TRIM to remove them.
- One side stored as text and the other as a number. Use VALUE or format both consistently.
- Case differences are not the issue (XLOOKUP is not case-sensitive), but hidden characters like non-breaking spaces are.
- The lookup array does not fully cover the data (wrong range).
- The value contains a similar but different character, for example a slash instead of a hyphen.

A quick diagnostic: =A2=B2 to check if the two values are literally equal, and LEN() on each to check character counts.

QUESTION 11

What is OFFSET and where is it used?

ANSWER

OFFSET returns a cell or range that is a given number of rows and columns from a starting cell. It is used to build dynamic ranges that change size, such as the last N months of data.

```
=OFFSET(A1, 0, 0, COUNTA(A:A), 1)  dynamic range covering all values in column A
```

OFFSET is volatile (it recalculates on every change) so it can slow down large workbooks. INDEX is a non-volatile alternative for building dynamic ranges.

QUESTION 12

What is INDIRECT and when should you use it carefully?

ANSWER

INDIRECT converts a text string into a valid cell reference. It is used to build formulas whose target range depends on another cell's value, for example switching between sheets.

```
=SUM(INDIRECT("'"&A1&"'!B2:B10"))  sums B2:B10 on the sheet named in A1
```

INDIRECT is volatile and does not update when referenced sheets are renamed. Use it sparingly.

QUESTION 13

What does CHOOSE do?

ANSWER

CHOOSE picks one of many values based on an index number.

```
=CHOOSE(A1, "Low", "Medium", "High")
```

It is used for simple index-to-label conversions and inside array formulas to reshape ranges.

QUESTION 14

What is the difference between VLOOKUP and XLOOKUP when the lookup value is not found?

ANSWER

VLOOKUP returns #N/A. XLOOKUP returns whatever you specify in the fourth argument, so you do not need IFERROR:

```
=XLOOKUP(A2, IDs, Names, "Not Found")
```

This is cleaner, less nested and easier to maintain.

PRACTICAL APPLICATION

- Pulling employee names from a master sheet into a monthly attendance report using XLOOKUP.
- Looking up product prices in a pricing sheet using INDEX-MATCH so a new column can be inserted without breaking the formula.
- Building a two-way pay-grid lookup for HR reports using INDEX with two MATCHes.
- Using CHOOSE to map numeric status codes to friendly labels.

INTERVIEWER FOLLOW-UP

Q. What happens if VLOOKUP finds multiple matching values?

A. VLOOKUP always returns the first match from the top. To handle duplicates, use XLOOKUP with search_mode of -1 to search from bottom, use FILTER for multiple results, or use an INDEX-SMALL array formula.

Q. If you have Microsoft 365, why would you still learn INDEX-MATCH?

A. Because files are often shared across teams with mixed Excel versions. INDEX-MATCH works everywhere; XLOOKUP does not exist in older Excel and shows #NAME? there.

16

Date and Time Functions

WHAT YOU SHOULD KNOW

Excel stores dates as serial numbers (1 = 1 Jan 1900). Times are stored as fractions of a day (0.5 = 12:00 noon). Because dates are numbers, you can subtract them, sort them and use them in calculations. Most date interview questions test whether the candidate understands this and can handle real business date logic.

INTERVIEW QUESTIONS

QUESTION 01

How does Excel store dates internally?

ANSWER

Dates are stored as sequential serial numbers. The value 1 represents 1 January 1900. So 45000 represents a date about 123 years later. This is why you can add and subtract dates directly. It is also why a date typed as text will not sort or subtract correctly.

QUESTION 02

What is the difference between TODAY and NOW?

ANSWER

TODAY returns the current date only. NOW returns the current date and time. Both are volatile, so they update every time the workbook recalculates. Use TODAY for date differences, use NOW for timestamp logs.

QUESTION 03

How would you calculate the number of years an employee has worked with the company?

ANSWER

Use DATEDIF with the "Y" unit:

```
| =DATEDIF(HireDate, TODAY(), "Y")
```

DATEDIF returns the completed years, months or days between two dates depending on the unit ("Y", "M", "D", "YM", "MD", "YD"). It is not shown in the Excel function list but it works.

QUESTION 04

What is the difference between NETWORKDAYS and WORKDAY?

ANSWER

NETWORKDAYS returns the number of working days between two dates, excluding weekends and any holidays you provide. WORKDAY returns a date that is a given number of working days after or before a start date.

<code>=NETWORKDAYS(A2, B2, holidays)</code>	number of working days between two dates
<code>=WORKDAY(A2, 10, holidays)</code>	10 working days after A2, skipping holidays

Used in SLA tracking, project deadlines and payroll calculations.

QUESTION 05

How do you extract the month name from a date?

ANSWER

Use TEXT with a month format:

<code>=TEXT(A2, "mmmm")</code>	returns full month name like "January"
<code>=TEXT(A2, "mmm")</code>	returns short month like "Jan"

Do not use MONTH for this, because MONTH returns a number (1 to 12).

QUESTION 06

How would you find the last day of the month for any given date?

ANSWER

Use EOMONTH:

<code>=EOMONTH(A2, 0)</code>	last day of the same month
<code>=EOMONTH(A2, 1)</code>	last day of next month

EDATE works similarly for the same day of a future or past month.

QUESTION 07

How would you identify records where the due date has passed?

ANSWER

Compare the due date with TODAY:

<code>=IF(DueDate<TODAY(),"Overdue","On Time")</code>
--

For a Pivot Table friendly version, wrap it in a helper column called Status. This is common in AR (accounts receivable) reports.

QUESTION 08

How do you calculate age from a date of birth?

ANSWER

DATEDIF is the cleanest approach:

```
=DATEDIF(DOB, TODAY(), "Y")
```

For years and months together, use:

```
=DATEDIF(DOB, TODAY(), "Y") & " Y " & DATEDIF(DOB, TODAY(), "YM") & " M"
```

QUESTION 09

Why might a date column not sort correctly?

ANSWER

Usually because some of the dates are stored as text. Text dates and real dates in the same column sort inconsistently. Fixes:

- Select the column, use Text to Columns, and complete the wizard with Date format selected.
- Multiply by 1 or use DATEVALUE.
- Import the file through Power Query and set the column type to Date.

QUESTION 10

How do you calculate the difference in months or years between two dates?

ANSWER

Use DATEDIF:

```
=DATEDIF(start, end, "M")    completed months between dates
=DATEDIF(start, end, "Y")    completed years between dates
```

DATEDIF ignores partial periods; it returns whole units. If you need a decimal age, compute (end - start)/365.25 approximately.

PRACTICAL APPLICATION

- Building tenure reports for HR using DATEDIF from date of joining to today.
- Producing month-year labels for grouping in Pivot Tables using TEXT.
- Calculating SLA breach flags using NETWORKDAYS and TODAY.
- Grouping data by fiscal month or quarter using EOMONTH.

INTERVIEWER FOLLOW-UP

Q. Why does subtracting one date from another give a big number instead of a normal date?

A. Because dates are numbers internally. The result is the count of days, formatted by Excel as a date. Format the result cell as General or Number to see the correct count of days.

17

Text Functions

WHAT YOU SHOULD KNOW

Text functions are used almost entirely for data cleaning: standardizing names, splitting or combining fields, removing junk characters, and reformatting strings. Analysts spend a significant portion of their time on text cleaning, so interviewers test these functions heavily.

INTERVIEW QUESTIONS

QUESTION 01

What is the difference between LEFT, RIGHT and MID?

ANSWER

All three extract characters from a string. LEFT extracts from the beginning, RIGHT from the end, and MID from any starting position.

=LEFT("Data Analyst", 4)	returns "Data"
=RIGHT("Data Analyst", 7)	returns "Analyst"
=MID("Data Analyst", 6, 3)	returns "Ana"

QUESTION 02

What does LEN do?

ANSWER

LEN returns the number of characters in a text string. It is used to validate field lengths (for example a PAN card should be 10 characters), to check for hidden characters, and to build dynamic extractions when combined with FIND or SEARCH.

QUESTION 03

What is the difference between UPPER, LOWER and PROPER?

ANSWER

UPPER converts text to all capitals. LOWER converts it to all lowercase. PROPER capitalizes the first letter of each word. PROPER is used for standardizing customer names but should be checked, because it also capitalizes things like "Iphone" into "Iphone" which may be wrong.

QUESTION 04

What does TRIM do and where does it help?

ANSWER

TRIM removes leading and trailing spaces from text and reduces multiple spaces between words to one. It is used to fix data where users have accidentally added spaces. TRIM does not remove non-breaking spaces (character 160) that often come from web copy-paste; those need SUBSTITUTE or CLEAN.

QUESTION 05

What is the difference between FIND and SEARCH?

ANSWER

Both return the position of one text inside another. Two differences:

- FIND is case-sensitive. SEARCH is not.
- SEARCH supports wildcards (* and ?). FIND does not.

Use SEARCH for most business scenarios where case does not matter. Use FIND when case matters.

QUESTION 06

What is the difference between SUBSTITUTE and REPLACE?

ANSWER

SUBSTITUTE replaces specific text wherever it appears. REPLACE replaces a specific number of characters at a specific position.

=SUBSTITUTE("Mr. Sharma", "Mr.", "Mr")	returns "Mr Sharma"
=REPLACE("abc123", 4, 3, "XYZ")	returns "abcXYZ"

SUBSTITUTE is far more common in cleaning. REPLACE is used when the location is known but the content is variable.

QUESTION 07

What is the difference between CONCAT, CONCATENATE and TEXTJOIN?

ANSWER

CONCATENATE is the legacy function that joins text values. CONCAT is its modern replacement and also accepts ranges. TEXTJOIN is the most powerful: it joins values with a chosen delimiter and can skip empty cells.

=CONCAT(A2:D2)	joins all cells with no separator
=TEXTJOIN(",", TRUE, A2:D2)	joins with ",", " and skips blanks

Use TEXTJOIN for building comma-separated lists like tag columns or address lines.

QUESTION 08**How would you extract the first name from a full name column?****ANSWER**

If names are stored as "Rahul Sharma":

```
=LEFT(A2, FIND(" ", A2) - 1)
```

This finds the space and takes everything before it. For last name, use MID with FIND from the space onwards. In modern Excel you can also use TEXTBEFORE and TEXTAFTER, or Flash Fill if the transformation is one-off.

QUESTION 09**How would you clean a text column with extra spaces, mixed case and hidden characters?****ANSWER**

Build one formula in a helper column:

```
=PROPER(TRIM(CLEAN(SUBSTITUTE(A2, CHAR(160), " "))))
```

Order: SUBSTITUTE removes non-breaking spaces, CLEAN removes non-printable characters, TRIM removes extra spaces, PROPER standardizes case. Then paste the helper column back as values.

QUESTION 10**How do you convert a text stored number back into a real number?****ANSWER**

Several options:

- Use VALUE(A2).
- Multiply the cell by 1: =A2*1.
- Use Text to Columns and finish with the General or Number format selected.
- Paste Special > Multiply with 1 from another cell.

PRACTICAL APPLICATION

- Cleaning customer name lists before uploading to a CRM.
- Splitting an "Email" field into username and domain using FIND and MID.
- Building comma-separated tag columns for a product catalog using TEXTJOIN.
- Extracting the first three letters of a product code to identify the category.

18

Dynamic Array Functions

WHAT YOU SHOULD KNOW

Dynamic arrays are a newer feature in Excel 365 and Excel 2021. A single formula can return many results that spill into adjacent cells automatically. This replaces many array-formula tricks that older Excel needed. Key functions: FILTER, SORT, SORTBY, UNIQUE, SEQUENCE, RANDARRAY.

INTERVIEW QUESTIONS

QUESTION 01

What is a dynamic array in Excel?

ANSWER

A dynamic array is a formula that returns multiple values and automatically spills them into neighbouring cells. The formula is written in one cell and its output extends as needed. The output range is called the spill range, and its top-left cell shows the formula.

QUESTION 02

What does the FILTER function do?

ANSWER

FILTER returns rows from a range that meet one or more conditions. It replaces manual filtering when the result needs to be a live formula.

```
=FILTER(Data, Data[Region]="West", "No records")
```

The third argument is what to return if no rows match.

QUESTION 03

What does UNIQUE do?

ANSWER

UNIQUE returns a list of distinct values from a range. It replaces Remove Duplicates when the result should stay live.

```
=UNIQUE(A2:A1000)    distinct list of values from column A
```

Combined with FILTER, it can return unique values based on a condition.

QUESTION 04

What is the difference between SORT and SORTBY?

ANSWER

SORT sorts a range by one of its own columns. SORTBY sorts a range by one or more separate arrays.

=SORT(Data, 3, -1)	sort Data by 3rd column descending
=SORTBY(Names, Revenue, -1)	sort Names by Revenue descending

SORTBY is more flexible when the sort key is not part of the output range.

QUESTION 05

What is the SEQUENCE function used for?

ANSWER

SEQUENCE generates a list of numbers in a range or a matrix.

=SEQUENCE(10)	numbers 1 through 10
=SEQUENCE(5,2,10,5)	5 rows, 2 cols, starting at 10, step 5

It is used to build helper series for calculations, generate month numbers, or feed other formulas that need an incrementing input.

QUESTION 06

What is a spill error (#SPILL!)?

ANSWER

A #SPILL! error appears when a dynamic array cannot expand into its intended range because another value is blocking it. To fix it, remove the blocking value or move the formula to a location with enough empty space.

QUESTION 07

How would you get a unique list of customers who spent more than 10,000?

ANSWER

Combine FILTER and UNIQUE:

=UNIQUE(FILTER(Data[Customer], Data[Spend]>10000))
--

One formula, always live, no manual filtering. This kind of composition is one of the biggest strengths of dynamic arrays.

QUESTION 08

What is the # symbol in dynamic array formulas?

ANSWER

The # symbol refers to the whole spill range from a dynamic formula. If cell A1 holds a UNIQUE formula that spills down, A1# refers to the entire spilled list. It is used when you want to reference a spill range in another formula without knowing its exact size.

PRACTICAL APPLICATION

- Building a live filtered list of overdue invoices using FILTER.
- Generating a distinct customer list for a drop-down using UNIQUE.
- Ranking products by revenue using SORTBY on the fly.
- Producing a rolling 12-month index using SEQUENCE for chart plotting.

INTERVIEWER FOLLOW-UP

Q. Do dynamic array functions work in Excel 2016?

A. No. FILTER, SORT, UNIQUE, SEQUENCE and XLOOKUP require Microsoft 365 or Excel 2021. In older Excel you still need INDEX-MATCH, Advanced Filter, and array formulas entered with Ctrl+Shift+Enter.

19

Pivot Tables, Slicers and Timelines

WHAT YOU SHOULD KNOW

A Pivot Table summarizes large data into an interactive report by dragging fields into Rows, Columns, Values and Filters. It is one of the most important Excel skills for a Data Analyst. Slicers and Timelines are visual filter controls that make Pivot Tables interactive. Almost every Excel interview will include at least three Pivot Table questions.

INTERVIEW QUESTIONS

QUESTION 01

What is a Pivot Table and why is it used?

ANSWER

A Pivot Table is a summary report generated from raw data. It groups data by chosen fields and applies aggregations (sum, count, average and others) without writing formulas. It is used because it is fast, dynamic and much easier to change than manual formulas.

QUESTION 02

When would you use a Pivot Table instead of formulas?

ANSWER

Use a Pivot Table when the summary might change often, when the grouping fields might change, when the audience wants to slice the data themselves, or when the dataset is large enough that hundreds of SUMIFS would be slow and hard to maintain.

QUESTION 03

What are the four areas of a Pivot Table?

ANSWER

Rows: fields that appear down the left, defining categories. Columns: fields across the top, creating a cross-tab. Values: numeric fields that get aggregated. Filters: fields that filter the entire report from the top.

EXAMPLE :

For sales analysis: Region in Rows, Category in Columns, Revenue in Values, Year in Filters.

QUESTION 04**What happens when the source data of a Pivot Table changes?****ANSWER**

The Pivot Table does not update by itself. You must Refresh it (right-click the Pivot, choose Refresh). If new rows or columns were added outside the original range, the Pivot will not include them even after refresh unless the source is expanded, which is why converting the source to an Excel Table before creating the Pivot is a best practice.

QUESTION 05**Why might a Pivot Table not show newly added rows?****ANSWER**

Because the source range is fixed at the size when the Pivot was created. Solutions:

- Convert the source to an Excel Table before building the Pivot.
- Change the source range through Pivot Table Analyze > Change Data Source.
- Use a named range that grows dynamically.

QUESTION 06**How would you group dates in a Pivot Table by month and year?****ANSWER**

Right-click any date value in the Pivot > Group > choose Months and Years. Excel then lets you drill down by year and month. If grouping options are greyed out, the date column has mixed types (some dates stored as text). Clean the source column first.

QUESTION 07**What is a Calculated Field in a Pivot Table?****ANSWER**

A Calculated Field is a formula that uses existing fields to create a new metric inside the Pivot Table itself, without adding a column to the source data.

EXAMPLE:

If the source has Revenue and Cost, a Calculated Field named Profit can be defined as =Revenue - Cost. This new field shows up in the Values area.

QUESTION 08**What is the difference between a Calculated Field and a Calculated Item?****ANSWER**

A Calculated Field creates a new field using existing fields. A Calculated Item creates a new item within an existing field, useful when you want to combine categories, for example creating a

"Metro Cities" item that sums Mumbai + Delhi + Bangalore. Calculated Items can cause double counting so they should be used carefully.

QUESTION 09

What are Slicers and Timelines?

ANSWER

Slicers are visual filter buttons for any field. Timelines are visual filters specifically for date fields, showing a horizontal ribbon by day, month, quarter or year. Both make dashboards interactive without touching the Pivot Table field list.

QUESTION 10

How can one slicer control multiple Pivot Tables?

ANSWER

Right-click the slicer, choose Report Connections, and tick every Pivot Table you want it to control. All connected Pivots must share the same source data. This is the foundation of a Pivot-based dashboard, so selecting a region in one slicer filters every chart at once.

QUESTION 11

How would you calculate percentage of total in a Pivot Table?

ANSWER

Right-click the value field, choose Show Values As > % of Grand Total (or % of Row Total, % of Column Total, % of Parent Row). No formula needed. This is used to show contribution of each region, product or category to overall revenue.

QUESTION 12

What is the difference between a Pivot Table and Power Query?

ANSWER

Pivot Tables summarize data. Power Query transforms and prepares data. Power Query loads clean data; a Pivot Table then aggregates it. In practice, Power Query does the ETL and Pivot Tables do the reporting. They complement each other rather than compete.

QUESTION 13

How would you find the top 5 products by revenue in a Pivot Table?

ANSWER

Add Product to Rows and Revenue to Values. Then right-click a product, choose Filter > Top 10, and set the number to 5 by Sum of Revenue. Excel filters the Pivot to only the top 5. Sort descending for a clean leaderboard.

QUESTION 14**What are some limitations of Pivot Tables?****ANSWER**

Common ones to know:

- They aggregate; they cannot easily return row-level detail.
- Formatting is reset on refresh unless "Preserve cell formatting" is enabled.
- Multiple Pivot Tables based on independent sources can behave inconsistently on shared slicers.
- They do not automatically expand to new data unless the source is an Excel Table.
- Complex calculations may need Measures (Data Model / Power Pivot) instead of Calculated Fields.

PRACTICAL APPLICATION

- Building monthly revenue summaries by region and category.
- Ranking top and bottom performers using a Pivot Filter.
- Producing contribution percentage reports using Show Values As.
- Building slicer-driven executive dashboards.

INTERVIEWER FOLLOW-UP

Q. If two Pivots use different sources but you want a single slicer to filter both, what do you do?

A. Load both sources into the Data Model and connect them through a shared dimension (for example, a Region table). Then the slicer can filter both Pivots because they now share the same dimension in the model.

20

Power Query

WHAT YOU SHOULD KNOW

Power Query is Excel's built-in ETL tool. ETL stands for Extract, Transform, Load. Every step you take in Power Query is recorded, so the entire cleaning workflow can be refreshed with one click when the source file changes. This is the single biggest productivity tool a Data Analyst has in Excel.

INTERVIEW QUESTIONS

QUESTION 01

What is Power Query and why is it important for an analyst?

ANSWER

Power Query is an ETL engine inside Excel that connects to data sources, transforms the data through a series of recorded steps, and loads the result to a worksheet or Data Model. It is important because it turns manual repeated cleaning into a repeatable, refreshable process. What used to take 30 minutes every month can be done in one click.

QUESTION 02

What is meant by ETL?

ANSWER

ETL means Extract, Transform, Load. Extract: get the data from source files, databases or web pages. Transform: clean, reshape and enrich the data. Load: send the finished dataset to Excel, a Pivot Table, or the Data Model. Power Query is Excel's ETL layer.

QUESTION 03

What sources can Power Query connect to?

ANSWER

CSV and text files, Excel workbooks, folders (all files in a folder combined), SQL Server and other databases, SharePoint, web pages, JSON and XML, PDFs and more. In practice most analysts use file, folder and web connections.

QUESTION 04**What is the difference between Merge and Append in Power Query?****ANSWER**

Append stacks queries on top of each other, adding rows. It is used to combine multiple similar files, for example twelve monthly sales files into one. Merge joins queries side by side on a matching column, like a SQL JOIN. It is used to bring related data from another table, such as adding customer name to an order table.

QUESTION 05**What join types are available in Merge?****ANSWER**

The join options in Merge are:

- Left Outer: all rows from the first table, matching rows from the second.
- Right Outer: all rows from the second table.
- Full Outer: all rows from both.
- Inner: only rows where the key exists in both tables.
- Left Anti: rows in the first table that have no match in the second (useful for finding missing records).
- Right Anti: the opposite.

QUESTION 06**What is Pivot and Unpivot in Power Query?****ANSWER**

Pivot turns unique values from one column into multiple columns. Unpivot does the reverse: it turns columns into rows. Unpivot is especially important for cleaning data that arrives with months across columns instead of down rows, which is common in reports from finance and operations.

EXAMPLE :

If a source has columns Region, Jan, Feb, Mar, Apr..., Unpivot creates just three columns: Region, Month, Value. This is the format Pivot Tables and Power BI actually need.

QUESTION 07**How would you combine 12 monthly CSV files with identical columns using Power Query?****ANSWER**

Place all files in one folder. In Excel go to Data > Get Data > From File > From Folder and choose the folder. Power Query lists the files and lets you combine them using a sample file. It creates a query that reads every file in the folder, applies the same transformations, and stacks the results. When a 13th file is dropped into the folder, one refresh brings it in.

QUESTION 08

What is a Custom Column in Power Query?

ANSWER

A Custom Column is a new column added through a formula in M language, or through the Add Column menu. It is used to create derived fields like Profit = Revenue - Cost, or a Region Group based on a lookup, without editing the source data.

QUESTION 09

What is Data Profiling in Power Query?

ANSWER

Data Profiling shows quality indicators at the top of each column: empty values, errors, distinct and unique counts, and a mini-distribution chart. Turn it on from the View tab. It gives a quick health check before loading, so you catch nulls, blanks and outliers before they hit your report.

QUESTION 10

When would you use Power Query instead of formulas?

ANSWER

Use Power Query when:

- The same cleaning steps repeat every week or month.
- The data comes from an external file or folder.
- Multiple files need to be combined.
- The data needs reshaping (pivot, unpivot, merge).
- You want the transformations to be documented as an auditable list of steps.

Use formulas when the calculation is small, one-off, or must live in the sheet cells for a business user to see and edit.

QUESTION 11

When would you use Power Query instead of VBA?

ANSWER

Power Query is preferred for repeatable data preparation because it is safer, easier to hand over, easier to audit, and does not require coding. Use VBA for tasks Power Query cannot do:

interacting with the Excel UI, sending emails, showing dialogs, event-based automation, and bulk file operations that are not just about combining data.

QUESTION 12

What does Refresh do and how does it work with Power Query?

ANSWER

Refresh re-runs every step in the query against the current source data. Right-click a query and Refresh, or use Data > Refresh All. Refresh is what makes Power Query "repeatable": every future dataset flows through the same cleaning steps automatically.

QUESTION 13

What is a Query Dependency and why does it matter?

ANSWER

Queries can depend on other queries. For example, a "Cleaned Sales" query may depend on a "Raw Sales" query. Power Query has a Dependencies view that shows the chain visually. It matters because breaking the source query breaks everything downstream, and refresh order can affect performance.

QUESTION 14

Why can merging data in Power Query produce more rows than expected?

ANSWER

If the key column in the right table has duplicate values, each match generates a new row on the left. For example, joining Orders to Customers on Customer ID where a Customer ID appears twice in Customers will double the matching order rows. Always check that the key in the right table is unique, or aggregate the right table first.

PRACTICAL APPLICATION

- Consolidating 12 monthly sales CSV files into a single cleaned dataset that refreshes when new files land in the folder.
- Merging an Orders table with a Products table to bring the product category into the order line.
- Unpivoting a wide report (months as columns) into a tall format for Pivot Tables.
- Cleaning inconsistent city names using conditional columns and lookups within Power Query.

INTERVIEWER FOLLOW-UP

Q. A stakeholder asks why they should trust a Power Query workflow more than a manually cleaned file. What do you say?

A. Because every step is recorded and repeatable. A manual clean depends on whoever did it and cannot be re-run. A Power Query workflow can be audited by any analyst, produces the same result every month, and takes seconds to refresh when new data arrives.

21

Data Cleaning

WHAT YOU SHOULD KNOW

Data cleaning is where analysts spend a large share of their time. Real business data is rarely clean: names are inconsistent, numbers are stored as text, dates are broken, duplicates hide, and blanks are everywhere. This section covers realistic cleaning problems and how to solve them in Excel.

INTERVIEW QUESTIONS

QUESTION 01

You receive a customer dataset where names have inconsistent capitalization and extra spaces. How would you clean it?

ANSWER

Add a helper column with a combined formula:

```
=PROPER(TRIM(A2))
```

TRIM removes extra spaces, PROPER standardizes capitalization. Copy the helper column, paste as values back onto the original column, and delete the helper. If the data also has non-breaking spaces (from web copy), first wrap in SUBSTITUTE(A2, CHAR(160), " ").

QUESTION 02

How do you find and remove duplicate records?

ANSWER

Two approaches based on need:

- To remove: Data > Remove Duplicates, choose the columns that define a duplicate.
- To identify without deleting: Conditional Formatting > Highlight Cells Rules > Duplicate Values, or use =COUNTIF(range,cell)>1 in a helper column.

For a recurring workflow, do this in Power Query so it can be re-run every month.

QUESTION 03

How do you handle missing values in Excel?

ANSWER

The approach depends on business rules:

- For numeric columns, decide whether to leave blank, fill with zero, or fill with the mean or median.
- For text columns, fill with "Unknown" or a category-specific default.
- If missingness is random, the row can sometimes be dropped.
- If the row is critical (like Customer ID), flag it for correction rather than filling in a value.

Use Go To Special > Blanks to select all blank cells at once, then apply the fix.

QUESTION 04

A column has numbers stored as text. How do you convert them?

ANSWER

Any of the following:

- Use VALUE(A2) in a helper column.
- Multiply the column by 1: =A2*1.
- Text to Columns wizard, finish with the General format.
- Paste Special > Multiply with 1 from an empty cell.

Confirm the fix with =ISNUMBER(A2), which should return TRUE.

QUESTION 05

Dates in a column are stored as text like "10-Jan-2024". How do you convert them to real dates?

ANSWER

Use DATEVALUE in a helper column: =DATEVALUE(A2). Then format the helper as Date. For a bulk conversion, use Text to Columns and pick Date on the final step of the wizard, choosing the correct source order (DMY or MDY). For an automated repeat, do it in Power Query with a set-type step.

QUESTION 06

A column of city names has variations like "Mumbai", " Mumbai", "mumbai" and "MUMBAI". How do you standardize?

ANSWER

One combined formula handles most cases:

| =PROPER(TRIM(A2))

For controlled categories, add a lookup table with the "clean" spelling and use XLOOKUP or Power Query to map raw values to the standard version. This is safer than PROPER when city names are business codes.

QUESTION 07

How would you handle common Excel errors like #N/A, #VALUE!, #REF!, #DIV/0! and #NAME?

ANSWER

Each has a specific cause:

- #N/A: value not found in a lookup. Fix the source or wrap with IFNA.
- #VALUE!: wrong argument type, often text where a number is expected. Check inputs.
- #REF!: a reference has been deleted, usually by removing a column. Rebuild the formula.
- #DIV/0!: division by zero or by a blank cell. Guard with IFERROR or an IF check.
- #NAME?: unknown function or misspelled name. Check spelling and check that the function exists in the current Excel version.

Avoid hiding errors blindly with IFERROR; find and fix the cause first.

QUESTION 08

How do you standardize inconsistent categories like "USA", "U.S.A" and "United States"?

ANSWER

Create a mapping table with two columns: Raw Value and Standard Value. Add a helper column in the main data using XLOOKUP or VLOOKUP against the mapping table. For automated repeat, do the same in Power Query. This is safer than nested IFs because adding new variations only means adding a row to the mapping.

QUESTION 09

What is the difference between a blank cell and a cell with a zero, and why does it matter?

ANSWER

A blank cell has nothing in it. A zero cell has the number 0. They look similar but behave differently: blanks are counted by COUNTBLANK, zeros are counted by COUNT. Charts often plot zeros as a point on the axis while blanks may be treated as a gap. Some formulas like AVERAGE ignore blanks but include zeros, which changes the result.

QUESTION 10

How would you detect leading or trailing spaces that are invisible to the eye?

ANSWER

Use LEN in a helper column to check character count against expected. Or use TRIM in a helper and compare to the original with =A2=TRIM(A2). If FALSE, the original has invisible spaces. Fix with TRIM or with Find & Replace using a space in the search box.

PRACTICAL APPLICATION

- Preparing a customer master file for a CRM upload where names, phone numbers and cities must be standardized.
- Cleaning a monthly sales export where numeric columns arrive as text and dates are inconsistent.
- Detecting duplicate invoice numbers before running an accounts payable payment batch.
- Standardizing product categories before feeding a Pivot Table dashboard.

22

Data Visualization and Charts

WHAT YOU SHOULD KNOW

A chart is a communication tool, not a decoration. Choose the chart type based on the question you are answering. A wrong chart hides the story even if the data is correct. Interviewers will often ask you to pick a chart for a specific business situation.

INTERVIEW QUESTIONS

QUESTION 01

What are the principles of a good chart?

ANSWER

Keep it simple. Match the chart type to the question. Use minimal colors. Label axes and units. Avoid 3D effects. Sort data meaningfully (largest to smallest for categorical). Highlight what matters and mute the rest.

QUESTION 02

When would you use a bar chart versus a column chart?

ANSWER

Both compare categories. Use a column chart (vertical) when there are few categories and the labels are short. Use a bar chart (horizontal) when there are many categories or the category labels are long, because they read better sideways.

QUESTION 03

When should you use a line chart?

ANSWER

When showing values over a continuous time period, especially when trend is more important than exact values. Multiple lines can compare series such as revenue across three years.

QUESTION 04

When should you avoid a pie chart?

ANSWER

Avoid pie charts when there are more than four or five slices, when the slices are similar in size, or when precise comparison is needed. Human eyes are poor at judging angles. A bar chart usually communicates the same data more clearly. Use pie charts only for a few high-contrast parts of a whole.

QUESTION 05

How would you visualize actual sales versus target?

ANSWER

A clustered column chart with two bars per period works. A better choice is a bullet chart or a combo chart where actuals are columns and target is a thin line. This keeps the target visible without dominating the visual.

QUESTION 06

How would you show the percentage contribution of each region to total revenue?

ANSWER

For a small number of regions, a pie or donut chart is acceptable. For more than five, use a stacked bar or a simple bar chart sorted by percentage. Show the percentages as labels so the eye does not have to estimate.

QUESTION 07

What is a combo chart and when is it used?

ANSWER

A combo chart combines two chart types on the same axes, most often a column for one series and a line for another. It is used when the two series have different scales or roles, for example revenue as columns and margin percentage as a line on a secondary axis.

QUESTION 08

What is a waterfall chart used for?

ANSWER

A waterfall chart shows how a starting value grows or shrinks through a series of positive and negative changes to reach an ending value. It is used in finance to walk from opening revenue to closing revenue through additions and deductions, or from budget to actual through the drivers of the variance.

QUESTION 09

When would you use a scatter plot?

ANSWER

When exploring the relationship between two continuous variables, such as ad spend versus revenue, or discount percentage versus units sold. If a pattern exists, the scatter reveals it visually.

QUESTION 10

What makes a dashboard difficult to understand?

ANSWER

Common problems:

- Too many colors, chart types and fonts.
- Charts that do not answer any specific question.
- Overlapping information, no visual hierarchy.
- Missing context (no title, unit or period).
- 3D effects and decoration that hide the numbers.

A good dashboard answers three to five business questions clearly and lets the user drill down when needed.

PRACTICAL APPLICATION

- Choosing a column chart for monthly sales and a line chart for cumulative sales.
- Using a waterfall chart to explain the variance between budget and actual.
- Using a scatter plot to test whether discount level relates to revenue growth.

23

Dashboards

WHAT YOU SHOULD KNOW

A dashboard is a one-view summary that helps a specific audience make specific decisions. Building a good dashboard is more about business thinking than Excel skill. Interviewers frequently test whether you can gather requirements and prioritize KPIs before touching Excel.

INTERVIEW QUESTIONS

QUESTION 01

What questions would you ask a manager before starting to build a sales dashboard?

ANSWER

A short list that shows analyst maturity:

- Who is the audience: leadership, sales managers, or reps?
- What decisions will this dashboard support?
- What are the three to five most important KPIs?
- What time granularity is needed: day, week, month, quarter?
- What filters do users need: region, product, team?
- How often will the data refresh, and from what source?
- Will this be viewed on a desktop, a projector, or mobile?

Answering these prevents building a dashboard that nobody uses.

QUESTION 02

What is a KPI and how do you choose one?

ANSWER

A Key Performance Indicator is a metric that measures progress toward a business goal. It should be measurable, tied to a decision, and few in number. Pick KPIs by working backwards from the question the audience wants to answer, not by listing every metric you can compute.

QUESTION 03**What is the difference between a KPI and a metric?****ANSWER**

Every KPI is a metric, but not every metric is a KPI. A metric is any number you track. A KPI is a metric tied to a strategic objective. Website visits is a metric. Website visits from a target country during a campaign week is a KPI.

QUESTION 04**What are common layout principles for an Excel dashboard?****ANSWER**

Place KPI cards at the top for immediate attention. Put filters and slicers near the top or on the left. Arrange charts in a logical reading order (top-left to bottom-right). Keep white space around each chart. Use consistent fonts and colors. Every element should earn its place.

QUESTION 05**How would you make a dashboard interactive in Excel?****ANSWER**

Use Pivot Tables as the calculation engine. Add slicers and timelines connected to the Pivots. Use form controls (drop-downs, option buttons) that write to a helper cell used by formulas. Use dynamic array formulas like FILTER for live filtering. For advanced cases, use the Data Model with measures.

QUESTION 06**How do you keep an Excel dashboard fast when the underlying data is large?****ANSWER**

Practical steps:

- Load the raw data through Power Query into the Data Model, not into a worksheet.
- Avoid volatile functions like OFFSET and INDIRECT in busy areas.
- Limit conditional formatting to necessary ranges.
- Reduce the number of formulas that recalculate on every action.
- Use Pivot Table calculations instead of dragging SUMIFS across thousands of cells.

QUESTION 07**How would you explain a dashboard insight to a non-technical stakeholder?****ANSWER**

Start with the finding in plain business language ("revenue in the West region grew 18% quarter on quarter, driven mainly by two products"). Then show the chart. Then explain the driver in one sentence. Save methodology and formulas for a technical appendix, not the executive conversation.

QUESTION 08

What is a KPI card in a dashboard?

ANSWER

A KPI card is a small visual block that shows one important number and its supporting context, for example current revenue, versus target, with a small color-coded status indicator. Multiple KPI cards at the top of a dashboard give the audience a snapshot before they look at the detailed charts.

PRACTICAL APPLICATION

- A sales dashboard with revenue, achievement %, top products, monthly trend, and region breakdown.
- An HR dashboard with headcount, attrition rate, hires, and department distribution.
- An operations dashboard with SLA compliance %, ticket backlog, and average handling time.

INTERVIEWER FOLLOW-UP

Q. A stakeholder asks for "one dashboard with everything". How do you respond?

A. Push back politely. A dashboard that shows everything shows nothing. Ask what decisions they want to make and design a focused view for that. If truly needed, offer a small suite of connected dashboards, each with a single purpose.

24

What-If and Scenario Analysis

WHAT YOU SHOULD KNOW

Excel provides three What-If tools: Goal Seek, Data Tables and Scenario Manager. They help answer questions about how outputs change when inputs change, without rewriting the model each time.

INTERVIEW QUESTIONS

QUESTION 01

What is What-If analysis in Excel?

ANSWER

What-If analysis is a set of tools that let you change input values and see the effect on a calculated result. It is used for pricing decisions, break-even analysis, forecast sensitivity and budgeting scenarios.

QUESTION 02

What is Goal Seek and when do you use it?

ANSWER

Goal Seek finds the input value needed to produce a target output. You specify the formula cell, the target value, and the cell to change.

EXAMPLE :

If $\text{profit} = \text{revenue} - \text{cost}$, and you want to know what revenue is required to reach a profit of 1 crore, Goal Seek adjusts the revenue cell until profit equals your target.

QUESTION 03

What is a Data Table in Excel and how is it different from a regular table?

ANSWER

A Data Table (Data > What-If Analysis > Data Table) computes a formula across a range of input values in one or two dimensions. It is not the same as an Excel Table. A one-variable Data Table shows results for one changing input; a two-variable Data Table shows results for combinations of two inputs, useful for sensitivity analysis.

QUESTION 04

What is Scenario Manager?

ANSWER

Scenario Manager stores named sets of input values (best case, base case, worst case) and lets you switch between them or produce a summary report comparing outputs. It is used when a model has more than two changing inputs, which is where Data Tables run out.

QUESTION 05

How would you build a sensitivity analysis for price and cost together?

ANSWER

Use a two-variable Data Table. Put price values across the top row, cost values down the left column, and reference the profit formula in the top-left cell. The result grid shows profit for every combination of price and cost, so you can see how sensitive profit is to changes in each.

PRACTICAL APPLICATION

- Using Goal Seek to find the discount percentage that keeps margin above 20%.
- Using a Data Table to test how monthly EMI changes with interest rate and tenure.
- Using Scenario Manager to compare best, base and worst quarterly forecasts.

25

Macros and VBA

WHAT YOU SHOULD KNOW

A macro is a saved set of Excel actions that can be replayed on demand. VBA (Visual Basic for Applications) is the programming language behind macros. For a Data Analyst role, deep VBA is rarely required, but understanding the basics is expected because most workplaces still have legacy VBA tools. When Power Query solves a problem, prefer it over VBA.

INTERVIEW QUESTIONS

QUESTION 01

What is a macro?

ANSWER

A macro is a recorded or written sequence of Excel actions that can be replayed with a shortcut, button or event. It is used to automate repetitive tasks such as formatting a report or refreshing multiple queries.

QUESTION 02

What is the difference between recording a macro and writing VBA code?

ANSWER

Recording captures your actions and generates VBA code automatically. It is quick and beginner friendly but the generated code is often long, uses Select statements, and does not handle variations. Writing VBA directly gives you loops, variables, error handling and much greater control. Analysts often record first, then clean up the code.

QUESTION 03

What is a Module in VBA?

ANSWER

A Module is a container inside a workbook where VBA procedures live. You open the VBA editor with Alt + F11, insert a new Module, and write Sub or Function procedures inside it. Modules keep code organized and separate from the workbook and worksheet objects.

QUESTION 04

What is the difference between a Sub and a Function in VBA?

ANSWER

A Sub performs an action but does not return a value. A Function performs an action and returns a value that can be used in another expression or as a custom worksheet formula.

```
Sub CleanData()    ... End Sub
Function AddTax(price, rate) ... End Function
```

QUESTION 05

What are variables and data types in VBA?

ANSWER

Variables are named storage for values. Data types include Integer, Long, Double, String, Boolean, Date, Variant, Range and Object. Declaring a type (Dim x As Long) makes the code faster, safer and easier to debug.

QUESTION 06

What are the main loop types in VBA?

ANSWER

For, For Each, Do While, Do Until. For loops through a number range. For Each loops through a collection (like every cell in a range). Do While and Do Until repeat while or until a condition holds. Choose based on whether the number of iterations is known in advance.

QUESTION 07

What are Workbook, Worksheet and Range objects?

ANSWER

These are the core Excel objects VBA works with. Workbook represents an open Excel file. Worksheet represents a tab inside a workbook. Range represents a cell or group of cells. Almost every VBA task operates on these three.

QUESTION 08

What is error handling in VBA?

ANSWER

Error handling stops your macro from crashing when something unexpected happens. The main structure is:

```
On Error GoTo ErrHandler
... code ...
Exit Sub
```

```
ErrHandler: MsgBox Err.Description
```

You can also use On Error Resume Next for tolerated errors, but this should be used carefully because it hides real problems.

QUESTION 09

When would you prefer Power Query over VBA?

ANSWER

For anything involving data preparation from files, folders, databases, or web sources: import, cleaning, merging, appending, unpivoting, and refresh. Power Query is safer, easier to hand off, and does not need programming. Use VBA for tasks Power Query cannot do, such as UI automation, email sending, event handling, and complex file operations.

QUESTION 10

What are macro security settings and why do they exist?

ANSWER

Macros can run arbitrary code, which is a security risk if the file comes from an untrusted source. Excel offers levels from "Disable all macros without notification" to "Enable all macros". The recommended setting is to disable macros with notification, so you can review the source before enabling. For trusted files, use Trusted Locations.

PRACTICAL APPLICATION

- Automating a monthly report by recording steps and cleaning up the code into a reusable macro.
- Looping through 50 files in a folder to extract a specific sheet from each.
- Building a small userform for data entry that writes to a worksheet.
- Automating an Outlook email dispatch with the final report attached.

INTERVIEWER FOLLOW-UP

Q. If a business user is scared of VBA, how do you decide whether to use it?

A. If Power Query can do the job, choose Power Query, since it is easier to hand off and maintain. Use VBA only when the task genuinely requires it, and document the code clearly with comments so another analyst can pick it up later.

SECTION 02

...

Practical Interview Questions

Now stop reading. Start solving.

26

Formula-Based Problems

Question 1: Calculate Revenue, Profit and Margin per Row

WHAT IS EXPECTED:

You have a table with columns Product, Units, Price, Cost. Calculate Revenue, Profit and Margin % for each row.

APPROACH:

Revenue is Units multiplied by Price. Profit is Revenue minus (Units times Cost). Margin is Profit divided by Revenue. Guard the margin formula against zero revenue.

FORMULA / EXCEL METHOD:

<code>=B2*C2</code>	Revenue
<code>=B2*C2 - B2*D2</code>	Profit
<code>=IFERROR((B2*(C2-D2))/(B2*C2),0)</code>	Margin %

EXPECTED OUTPUT:

Three new columns showing per-row Revenue, Profit and Margin percentage.

INTERVIEW FOLLOW-UP:

Q. If Price and Cost were both zero for a product, what would your Margin cell show?

Question 2: Calculate Achievement Percentage vs Target

WHAT IS EXPECTED:

You have Actual Sales and Target columns for each sales rep. Compute achievement % and flag reps who missed target.

APPROACH:

Achievement is Actual divided by Target. Flag column uses IF. Guard for zero target.

FORMULA / EXCEL METHOD:

<code>=IFERROR(Actual/Target, 0)</code>	Achievement %
<code>=IF(Actual>=Target, "Achieved", "Missed")</code>	Flag

EXPECTED OUTPUT:

Achievement column shows a percentage, Flag column shows Achieved or Missed for each rep.

INTERVIEW FOLLOW-UP:

Q. If Target is blank instead of zero, what happens to your formula?

Question 3: Calculate Month-over-Month Growth Percentage

WHAT IS EXPECTED:

You have a monthly revenue column. Compute the growth percentage vs the previous month.

APPROACH:

Growth is (This Month minus Last Month) divided by Last Month. First row is blank because there is no previous value. Guard for zero.

FORMULA / EXCEL METHOD:

`=IFERROR((B3 - B2)/B2, "")` entered from row 3 onwards

EXPECTED OUTPUT:

A column showing MoM growth as a percentage from the second month onwards.

INTERVIEW FOLLOW-UP:

Q. If a month has zero revenue and the next month has non-zero revenue, what does your formula return?

Question 4: Conditional Aggregation of Revenue

WHAT IS EXPECTED:

You have a table of transactions with columns Region, Category, Revenue. Compute total revenue for the West region and the Laptop category.

APPROACH:

Use SUMIFS with two criteria.

FORMULA / EXCEL METHOD:

`=SUMIFS(Revenue, Region, "West", Category, "Laptop")`

EXPECTED OUTPUT:

A single value: total revenue where region is West and category is Laptop.

INTERVIEW FOLLOW-UP:

Q. How would you adapt this to compute revenue for the West region only, ignoring category?

Question 5: Rank Sales Reps by Revenue

WHAT IS EXPECTED:

Given a list of reps and their revenue, produce a rank column with 1 being highest.

APPROACH:

Use RANK.EQ. Handle ties depending on business preference (RANK.AVG averages the ranks).

FORMULA / EXCEL METHOD:

`=RANK.EQ(B2, $B$2:$B$100, 0)`

EXPECTED OUTPUT:

A rank column with 1 for the highest revenue rep, 2 for the next, and so on.

INTERVIEW FOLLOW-UP:

Q. If two reps tie at revenue 100,000, what rank does each get?

Question 6: Percentage Contribution of Each Product**WHAT IS EXPECTED:**

A product-wise revenue list. Compute what percentage of total each product represents.

APPROACH:

Divide each product revenue by the grand total. Absolute reference on the total.

FORMULA / EXCEL METHOD:

```
=B2/SUM($B$2:$B$100)
```

EXPECTED OUTPUT:

A percentage column that adds up to 100% across all products.

INTERVIEW FOLLOW-UP:

Q. How would you achieve the same result inside a Pivot Table without a formula?

Question 7: Assign Grade Based on Score**WHAT IS EXPECTED:**

You have a Score column. Assign grade A for 90 and above, B for 75-89, C for 60-74, else D.

APPROACH:

Use IFS for readability; nested IF also works.

FORMULA / EXCEL METHOD:

```
=IFS(B2>=90, "A", B2>=75, "B", B2>=60, "C", TRUE, "D")
```

EXPECTED OUTPUT:

A grade column showing A, B, C or D per row.

INTERVIEW FOLLOW-UP:

Q. If you had to allow the bands to be edited by a business user without changing the formula, how would you redesign this?

27

Lookup Problems

Question 1: Find Employee Department

WHAT IS EXPECTED:

You have an EmployeeMaster table with Employee ID, Name and Department. In a monthly attendance sheet, add a Department column based on Employee ID.

APPROACH:

Use XLOOKUP or VLOOKUP against the master table.

FORMULA / EXCEL METHOD:

```
=XLOOKUP(A2, EmployeeMaster[Employee ID], EmployeeMaster[Department], "Not Found")
=VLOOKUP(A2, EmployeeMaster, 3, FALSE)    using VLOOKUP
```

EXPECTED OUTPUT:

A Department column populated from the master. Missing IDs show "Not Found" instead of #N/A.

INTERVIEW FOLLOW-UP:

Q. Why is XLOOKUP safer here than VLOOKUP if you later insert a new column into EmployeeMaster?

Question 2: Retrieve Product Price with Fallback

WHAT IS EXPECTED:

A PriceList table has Product Code and Price. In an orders sheet, pull the price for each order. If the code is not found, show "Price Missing".

APPROACH:

Use XLOOKUP with a not-found argument, or wrap VLOOKUP in IFNA.

FORMULA / EXCEL METHOD:

```
=XLOOKUP(A2, PriceList[Code], PriceList[Price], "Price Missing")
=IFNA(VLOOKUP(A2, PriceList, 2, FALSE), "Price Missing")
```

EXPECTED OUTPUT:

A Price column with numeric prices or the text "Price Missing" for unmatched codes.

INTERVIEW FOLLOW-UP:

Q. If a code appears multiple times in PriceList, which price does XLOOKUP return?

Question 3: Two-Criteria Lookup for Customer Discount

WHAT IS EXPECTED:

A Discount table has Customer Segment and Product Category as the two dimensions and the discount rate as the value. Given an order with Segment and Category, return the correct discount.

APPROACH:

Concatenate keys with XLOOKUP, or use INDEX with two MATCHes.

FORMULA / EXCEL METHOD:

```
=XLOOKUP(Segment & Category, Discount[Segment] & Discount[Category],
Discount[Rate], 0)

=INDEX(DiscountMatrix, MATCH(Segment, Segments, 0), MATCH(Category, Categories,
0))
```

EXPECTED OUTPUT:

A discount rate that reflects both the customer segment and the product category.

INTERVIEW FOLLOW-UP:

Q. What happens if a Segment-Category combination is missing from the discount table?

Question 4: Lookup Across Sheets

WHAT IS EXPECTED:

CustomerData is on Sheet1. Your report is on Sheet2. Bring the Customer Name into the report using Customer ID.

APPROACH:

Reference the sheet in the lookup range. Use XLOOKUP or VLOOKUP.

FORMULA / EXCEL METHOD:

```
=XLOOKUP(A2, Sheet1!A:A, Sheet1!B:B, "")

=VLOOKUP(A2, Sheet1!A:B, 2, FALSE)
```

EXPECTED OUTPUT:

Customer Name column populated from Sheet1 into the report.

INTERVIEW FOLLOW-UP:

Q. If Sheet1 is renamed to Customers, what happens to your formula?

Question 5: Duplicate Lookup Values

WHAT IS EXPECTED:

A ticket log has multiple entries per customer. You want the latest ticket status for each customer.

APPROACH:

Sort the log by date descending, then use XLOOKUP with search_mode -1 to search from bottom (or top, depending on how the data is sorted). For older Excel, sort and use INDEX-MATCH with LOOKUP tricks or aggregate in Power Query.

FORMULA / EXCEL METHOD:

```
=XLOOKUP(CustID, Log[CustID], Log[Status], "None", 0, -1)
```

EXPECTED OUTPUT:

Each customer shows the most recent status.

INTERVIEW FOLLOW-UP:

Q. Would you solve this with a Pivot Table? If yes, how?

Question 6: Handle Missing Lookup Value**WHAT IS EXPECTED:**

Your VLOOKUP is returning #N/A for many rows. Business wants "New Customer" instead.

APPROACH:

Wrap the VLOOKUP in IFNA (not IFERROR, so real formula errors are still visible).

FORMULA / EXCEL METHOD:

```
=IFNA(VLOOKUP(A2, CustMaster, 2, FALSE), "New Customer")
```

EXPECTED OUTPUT:

Column shows the mapped value where a match exists, and "New Customer" where it does not.

INTERVIEW FOLLOW-UP:

Q. Why prefer IFNA here over IFERROR?

28

Data Cleaning Problems

Question 1: Clean Names with Extra Spaces and Mixed Case

WHAT IS EXPECTED:

A customer name column has values like " rahul sharma", "RAHUL SHARMA" and "Rahul Sharma". Standardize all of them.

APPROACH:

Combine TRIM to remove extra spaces and PROPER to standardize case. Optionally SUBSTITUTE non-breaking spaces first.

FORMULA / EXCEL METHOD:

```
=PROPER(TRIM(SUBSTITUTE(A2, CHAR(160), " ")))
```

EXPECTED OUTPUT:

All names in Title Case with single spaces.

INTERVIEW FOLLOW-UP:

Q. What limitation does PROPER have when names contain something like McDonald?

Question 2: Remove Duplicate Records

WHAT IS EXPECTED:

A transactions table has duplicate rows because the same file was pasted twice. Remove them.

APPROACH:

Use Data > Remove Duplicates, selecting the columns that define a duplicate (usually all columns or a business key like Transaction ID). For a repeatable workflow, use Power Query.

FORMULA / EXCEL METHOD:

- Step 1: Convert range to Excel Table.
- Step 2: Data > Remove Duplicates > select all columns.
- Step 3: Confirm and verify the row count.

EXPECTED OUTPUT:

A dataset with only unique transaction rows.

INTERVIEW FOLLOW-UP:

Q. What is the difference between removing duplicates in Excel directly versus in Power Query?

Question 3: Standardize Category Values

WHAT IS EXPECTED:

A Region column has "USA", "U.S.A", "United States", and "us". Map them all to "United States".

APPROACH:

Create a Mapping table with Raw and Standard columns. Use XLOOKUP against the mapping. Load once, then reuse.

FORMULA / EXCEL METHOD:

```
=XLOOKUP(A2, Mapping[Raw], Mapping[Standard], A2)
```

EXPECTED OUTPUT:

A Standardized Region column with consistent values.

INTERVIEW FOLLOW-UP:

Q. How would you handle new variations that appear next month without editing the formula?

Question 4: Convert Text to Numbers

WHAT IS EXPECTED:

A Revenue column shows numbers left-aligned and SUM returns wrong totals. Convert to real numbers.

APPROACH:

Use VALUE, or multiply by 1, or run Text to Columns.

FORMULA / EXCEL METHOD:

```
=VALUE(A2)
=A2*1
```

Text to Columns > choose General on the final step.

EXPECTED OUTPUT:

Revenue column now right-aligned and included in SUM.

INTERVIEW FOLLOW-UP:

Q. How would you verify the fix worked?

Question 5: Fix Date Column Stored as Text

WHAT IS EXPECTED:

Dates are stored as text like "05-Jan-2024" and cannot be sorted correctly.

APPROACH:

Use DATEVALUE in a helper column, or use Text to Columns with the Date option and correct source order.

FORMULA / EXCEL METHOD:

```
=DATEVALUE(A2)
```

Then format the helper cell as Date.

EXPECTED OUTPUT:

A date column that sorts chronologically and can be used in date math.

INTERVIEW FOLLOW-UP:

Q. If some cells convert and others show #VALUE!, what is likely the cause?

Question 6: Extract First Name from Full Name

WHAT IS EXPECTED:

Full Name column has "Rahul Sharma". Extract first name into a new column.

APPROACH:

Use LEFT with FIND to locate the space, or TEXTBEFORE in modern Excel.

FORMULA / EXCEL METHOD:

```
=LEFT(A2, FIND(" ", A2) - 1)  
=TEXTBEFORE(A2, " ")    modern Excel
```

EXPECTED OUTPUT:

A First Name column.

INTERVIEW FOLLOW-UP:

Q. What happens if a cell contains only one name without a space?

29

Date and Time Problems

Question 1: Calculate Employee Tenure

WHAT IS EXPECTED:

You have Date of Joining. Compute tenure in years and months as of today.

APPROACH:

Use DATEDIF for whole units.

FORMULA / EXCEL METHOD:

```
=DATEDIF(A2, TODAY(), "Y") & " Y " & DATEDIF(A2, TODAY(), "YM") & " M"
```

EXPECTED OUTPUT:

A tenure string like "3 Y 5 M" for each employee.

INTERVIEW FOLLOW-UP:

Q. Why is DATEDIF not visible in the function library?

Question 2: Days Between Two Dates

WHAT IS EXPECTED:

Compute the number of days between an Order Date and a Delivery Date.

APPROACH:

Subtract dates directly, format the result as a number.

FORMULA / EXCEL METHOD:

```
=B2 - A2
```

EXPECTED OUTPUT:

A numeric column showing days difference.

INTERVIEW FOLLOW-UP:

Q. If the result appears as a date rather than a number, what setting do you change?

Question 3: Get Month-Year Label for Grouping

WHAT IS EXPECTED:

You have a Date column. Create a "Jan 2024" label for Pivot Table grouping.

APPROACH:

Use TEXT to format the date.

FORMULA / EXCEL METHOD:

```
=TEXT(A2, "mmm yyyy")
```

EXPECTED OUTPUT:

A text column with month-year labels.

INTERVIEW FOLLOW-UP:

Q. Would sorting this label alphabetically give correct chronological order?

Question 4: Working Days Between Two Dates

WHAT IS EXPECTED:

Compute the number of working days between two dates, excluding a holiday list.

APPROACH:

Use NETWORKDAYS with the holidays argument.

FORMULA / EXCEL METHOD:

```
=NETWORKDAYS(A2, B2, HolidayList)
```

EXPECTED OUTPUT:

A count of working days for each row.

INTERVIEW FOLLOW-UP:

Q. How would you exclude Fridays and Saturdays instead of Saturdays and Sundays?

Question 5: Flag Overdue Invoices

WHAT IS EXPECTED:

You have Due Date and Payment Status columns. Flag rows where the invoice is unpaid and the due date has passed.

APPROACH:

Combine AND with TODAY.

FORMULA / EXCEL METHOD:

```
=IF(AND(A2<TODAY(), B2<>"Paid"), "Overdue", "OK")
```

EXPECTED OUTPUT:

A Status column showing Overdue or OK per invoice.

INTERVIEW FOLLOW-UP:

Q. Why prefer TODAY over hard-coding a date?

Question 6: Get Last Day of Month for a Fiscal Report

WHAT IS EXPECTED:

For each transaction, return the last day of that month for grouping.

APPROACH:

Use EOMONTH with offset 0.

FORMULA / EXCEL METHOD:

```
=EOMONTH(A2, 0)
```

EXPECTED OUTPUT:

A column of month-end dates.

INTERVIEW FOLLOW-UP:

Q. How would you get the last day of the previous month instead?

30

Pivot Table Problems

Question 1: Sales by Region

WHAT IS EXPECTED:

Summarize a transactions table to show total revenue by region.

APPROACH:

Convert source to an Excel Table. Insert Pivot Table. Region in Rows. Revenue in Values (Sum).

FORMULA / EXCEL METHOD:

- Step 1: Ctrl+T on source to create a table.
- Step 2: Insert > PivotTable.
- Step 3: Drag Region to Rows and Revenue to Values.

EXPECTED OUTPUT:

A two-column summary of total revenue per region.

INTERVIEW FOLLOW-UP:

Q. If a new region is added to the source next week, will it show up automatically?

Question 2: Monthly Revenue Trend

WHAT IS EXPECTED:

Show revenue by month from a daily transactions table.

APPROACH:

Put Date into Rows and group by Month. Revenue into Values.

FORMULA / EXCEL METHOD:

- Step 1: Drag Date to Rows.
- Step 2: Right-click a date > Group > select Months and Years.
- Step 3: Drag Revenue to Values.

EXPECTED OUTPUT:

A month-by-month total, with year as an outer grouping when data spans years.

INTERVIEW FOLLOW-UP:

Q. What do you do if the Group option is greyed out?

Question 3: Top 5 Products by Revenue

WHAT IS EXPECTED:

Show only the top 5 revenue-generating products.

APPROACH:

Use Value Filters > Top 10 (choose Top 5) inside the Pivot Table.

FORMULA / EXCEL METHOD:

- Step 1: Drag Product to Rows and Revenue to Values.
- Step 2: Right-click Product > Filter > Top 10 > Top 5 by Sum of Revenue.
- Step 3: Sort descending on Revenue.

EXPECTED OUTPUT:

A leaderboard of 5 products from highest to lowest revenue.

INTERVIEW FOLLOW-UP:

Q. How would you also show these top 5 as a bar chart from the same Pivot?

Question 4: Contribution Percentage per Category

WHAT IS EXPECTED:

Show each category and what percentage of total revenue it represents.

APPROACH:

Add Category to Rows, Revenue to Values twice: once as Sum, once with Show Values As > % of Grand Total.

FORMULA / EXCEL METHOD:

- Step 1: Drag Category to Rows.
- Step 2: Drag Revenue to Values twice.
- Step 3: On the second Revenue field, Show Values As > % of Grand Total.

EXPECTED OUTPUT:

A table with Category, Revenue, and Revenue as %.

INTERVIEW FOLLOW-UP:

Q. Would you use Grand Total or Column Total when computing the percentage?

Question 5: Interactive Sales Dashboard with Slicers

WHAT IS EXPECTED:

Add filters for Region and Product Category that control multiple Pivot Tables.

APPROACH:

Insert slicers on both fields and connect them to every Pivot via Report Connections.

FORMULA / EXCEL METHOD:

- Step 1: Insert a Slicer for Region and one for Category.

Step 2: Right-click each slicer > Report Connections > tick all Pivot Tables.

EXPECTED OUTPUT:

A dashboard where clicking a region or category updates every chart at once.

INTERVIEW FOLLOW-UP:

Q. Why must all connected Pivots share the same data source?

31

Power Query Problems

Question 1: Consolidate 12 Monthly CSV Files

WHAT IS EXPECTED:

Every month you receive a CSV file into the same folder. Build one report that includes all files, and refreshes when a new file arrives.

APPROACH:

Use Get Data > From Folder in Power Query. Combine files using the sample file. Apply transformations once, and let refresh handle new months.

FORMULA / EXCEL METHOD:

- Step 1: Data > Get Data > From File > From Folder > choose folder.
- Step 2: Combine & Transform.
- Step 3: Verify column types.
- Step 4: Load to a worksheet or the Data Model.

EXPECTED OUTPUT:

A single table containing all monthly transactions. Dropping a new file into the folder and clicking Refresh brings it in.

INTERVIEW FOLLOW-UP:

Q. What happens if one file next month has a slightly different column name?

Question 2: Merge Orders with Customer Master

WHAT IS EXPECTED:

Bring Customer Name and Segment from the Customer Master into the Orders table.

APPROACH:

Load both tables into Power Query. Use Merge Queries on Customer ID with a Left Outer join. Expand the customer columns.

FORMULA / EXCEL METHOD:

- Step 1: Load Orders and Customers.
- Step 2: Home > Merge Queries > choose Customer ID as key, Left Outer.
- Step 3: Expand Name and Segment columns.

Step 4: Load to worksheet.

EXPECTED OUTPUT:

The Orders table with two new columns: Customer Name and Segment.

INTERVIEW FOLLOW-UP:

Q. What happens if a Customer ID exists in Orders but not in Customer Master?

Question 3: Unpivot a Wide Report

WHAT IS EXPECTED:

A report has Region in one column and Jan through Dec as separate columns. Convert it to a tall format with Region, Month and Value.

APPROACH:

In Power Query, select the month columns and use Unpivot Columns. Rename the result columns.

FORMULA / EXCEL METHOD:

- Step 1: Select the twelve month columns.
- Step 2: Transform > Unpivot Columns.
- Step 3: Rename Attribute to Month and Value to Revenue.

EXPECTED OUTPUT:

A tall table with three columns: Region, Month, Revenue.

INTERVIEW FOLLOW-UP:

Q. Why is a tall format better for Pivot Tables and dashboards?

Question 4: Clean a Repeating Weekly Report

WHAT IS EXPECTED:

Every week you get the same report with extra header rows, blanks and inconsistent types. Automate the cleanup.

APPROACH:

Load the file in Power Query. Remove top rows, promote headers, remove blank rows, set data types. Save as a query so next weeks file goes through the same steps.

FORMULA / EXCEL METHOD:

- Step 1: Get Data > From Excel.
- Step 2: Remove Top Rows and Promote Headers.
- Step 3: Remove empty rows and change column types.
- Step 4: Close & Load. Next week, replace the file and Refresh.

EXPECTED OUTPUT:

A clean, load-ready table every week without manual cleaning.

INTERVIEW FOLLOW - UP :

Q. How would you handle a case where next weeks file has an extra column?

32

Dashboard Problems

Question 1: Design an End-to-End Sales Dashboard

WHAT IS EXPECTED:

You are given transaction-level sales data. The sales manager wants to monitor revenue, profit, target achievement, regional performance and monthly trends. Design an Excel dashboard end to end.

APPROACH:

Follow a full analyst workflow: gather requirements, prepare data, build calculations, and design the visual layout.

FORMULA / EXCEL METHOD:

- Step 1: Requirements. Confirm the KPIs, filters (Region, Category), and time granularity.
- Step 2: Data preparation. Convert source to a table. Clean in Power Query if repeated. Add helper columns (Profit, Achievement %).
- Step 3: Pivot Tables. One per view: revenue by region, top products, monthly trend, achievement vs target.
- Step 4: Charts. Column chart for regions, line for monthly trend, combo for actual vs target, bar for top products.
- Step 5: KPI cards. Big numbers at the top for total revenue, total profit, achievement %.
- Step 6: Slicers and Timelines. Region and Category slicers, Date timeline. Connect to all Pivots.
- Step 7: Layout. KPIs at the top, filters at the left, charts in a clean grid, consistent fonts and colors.
- Step 8: Testing. Refresh with new data. Confirm filters flow across the dashboard.

EXPECTED OUTPUT:

A one-page dashboard with KPI cards, four to six charts, and slicers, driven by a single source of clean data.

INTERVIEW FOLLOW-UP:

Q. If the sales manager later wants a mobile-friendly view, how would you adapt the design?

Question 2: Design an HR Attrition Dashboard

WHAT IS EXPECTED:

You have employee-level data with joining date, exit date (blank if active), department and salary. Design a dashboard to track attrition.

APPROACH:

Define KPIs first (headcount, exits this month, attrition %). Build Pivots by department and month. Add a slicer for department and a timeline for month.

FORMULA / EXCEL METHOD:

Step 1: Add helper columns: Active (blank exit date), Exit Month.

Step 2: KPIs at top: Headcount, YTD Exits, Attrition Rate.

Step 3: Charts: Attrition trend by month (line), Attrition by department (bar), Tenure distribution (histogram).

Step 4: Slicers for Department and Location.

EXPECTED OUTPUT:

A single-page HR dashboard showing current headcount, attrition trend, and department-wise attrition.

INTERVIEW FOLLOW-UP:

Q. If leadership asks "why is attrition high in one department?", how would you extend this dashboard to help answer that?

How to Answer Project Questions

Almost every Data Analyst interview includes a question about your projects. Interviewers want to see how you think, not just what you did. Use a simple framework so the answer stays structured and you do not ramble.

ANSWER FRAMEWORK

9. Business problem: what were you trying to solve?
10. Dataset: what data did you have, and where did it come from?
11. Data cleaning: what issues did you find and how did you handle them?
12. Analysis: what did you calculate or explore?
13. Excel features used: what tools did you rely on?
14. Key insights: what did the analysis reveal?
15. Business impact: what decision or action came out of it?

SAMPLE ANSWER

I worked on a retail sales analysis project as part of my learning. The business problem was to understand which regions and products were driving revenue and which were underperforming, so that the sales team could focus on the right areas.

The dataset had about fifty thousand transactions with columns like Date, Region, Category, Product, Units, Revenue and Cost. The raw file had duplicate rows, extra spaces in region names, some numbers stored as text, and a few missing dates. I used Power Query to clean it: removed duplicates, standardized region names using a mapping table, converted text-numbers to numeric type, and filled missing categories with "Unknown".

For analysis I built Pivot Tables to summarize revenue by region and product, and calculated margin and month-over-month growth using SUMIFS and helper columns. I used XLOOKUP to bring product categories into the transactions from a master sheet.

The main insight was that two regions were contributing nearly sixty percent of the revenue, and one region was under-performing specifically on premium products. I built a one-page dashboard with KPI cards, region and category slicers, and monthly trend charts so the manager could filter and drill down. The business takeaway was to focus marketing on the two premium products in the under-performing region.

DO AND DO NOT

- Do use real numbers. Vague "many rows and lots of insights" sounds fake.
- Do explain your reasoning. Interviewers want to see decisions, not just steps.
- Do prepare one project deeply rather than four projects shallowly.
- Do not exaggerate the business impact. Interviewers spot inflation quickly.
- Do not memorize a script. If the interviewer asks a follow-up, you should not lose your place.

Final Revision Checklist

Use this checklist in the last twenty-four hours before an interview. Read each item and confirm that you could explain or demonstrate it. If you hesitate on any item, revisit that topic before you walk in.

FUNDAMENTALS AND REFERENCING

- Explain the difference between relative, absolute and mixed cell references.
- Demonstrate the F4 shortcut.
- Explain why a number stored as text does not sum correctly.
- Explain when to use an Excel Table instead of a plain range.

FORMULAS AND FUNCTIONS

- Use SUMIFS, COUNTIFS and AVERAGEIFS with multiple criteria.
- Write a nested IF and its cleaner IFS version.
- Explain the difference between IFERROR and IFNA.
- Explain the difference between VLOOKUP, INDEX-MATCH and XLOOKUP.
- Troubleshoot a lookup that returns #N/A.
- Use DATEDIF, EOMONTH and NETWORKDAYS.
- Use TRIM, PROPER, LEFT, MID, RIGHT, FIND, SEARCH and TEXTJOIN.
- Use FILTER, UNIQUE, SORT and SEQUENCE.

DATA CLEANING

- Clean a messy text column with inconsistent case and extra spaces.
- Convert a text-stored number and a text-stored date to their proper types.
- Standardize inconsistent category values using a mapping table.
- Explain the difference between a blank cell and a zero.

PIVOT TABLES AND DASHBOARDS

- Build a Pivot Table and explain the four areas.
- Group dates by month and year in a Pivot.
- Add slicers and connect them to multiple Pivots.
- Add a Calculated Field.
- Show values as percent of Grand Total.

- Design a simple KPI dashboard layout with cards, filters and charts.

POWER QUERY

- Import a CSV through Power Query and set data types.
- Explain the difference between Merge and Append.
- Consolidate multiple files from a folder.
- Unpivot a wide report into a tall table.
- Explain when to prefer Power Query over formulas or VBA.

CHARTS AND VISUALIZATION

- Choose the right chart for a given business question.
- Explain when a pie chart is a poor choice.
- Build a combo chart showing actuals as columns and target as a line.

AUTOMATION AND BUSINESS COMMUNICATION

- Explain what a macro is and when to use one.
- Explain when to use Power Query instead of VBA.
- Explain a chart or insight in plain business language.
- Talk through your Excel project using a clear framework.

Last-Minute Interview Tips

Practical reminders for the day of the interview. Read once before you walk in.

- Do not memorize definitions word for word. Understand the concept so you can explain it in your own words.
- Use examples. A short concrete example is more convincing than a paragraph of theory.
- If you do not know something, say so. Interviewers respect honesty far more than bluffing.
- Think out loud while solving a practical problem. Interviewers care about your reasoning, not just the final formula.
- Before building any report, restate the requirement to confirm you understood it.
- Focus on accuracy first, then presentation.
- Know the limitations of the functions you use. Interviewers often ask about edge cases.
- Prefer the cleaner solution over the cleverest one. Simplicity is a virtue in analytics.
- Prepare one project deeply. Be ready for follow-up questions on data size, cleaning steps and business impact.
- Keep answers short and structured. A thirty to sixty second answer that lands well is better than a three minute answer that meanders.
- Practice speaking your answers out loud, not just reading them. Interviews are verbal.
- Carry a notebook. Sketching a table or dashboard layout while talking always helps.

A Closing Note

An Excel interview is not a memory test. It is a test of how you think about data, how you explain your work, and how you choose the right tool for a business question. Every answer in this book has been written so that you understand the reasoning, not just the syntax.

Practice the topics on real data, prepare one project in depth, and speak your answers out loud before the interview. Do this consistently and you will walk in with the confidence that comes from understanding, not memorization.

If a topic here helped you, or if you would like to share a question you were asked in an interview, I would like to hear from you. Feedback from readers is what makes future editions of this book better than this one.

Best of luck.

Nikhil Mishra
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About the Author

Nikhil Mishra is a Data Analyst and educator based in India. His work sits at the intersection of analytics, technology and education.

Across the last two years, Nikhil has mentored more than twenty thousand students through online and offline cohorts, and has worked one-on-one with more than fifty who went on to land their first Data Analyst roles. His teaching draws directly from work experience: analytics fundamentals, data preparation, reporting and dashboards, along with practical exposure to data engineering concepts and cloud technologies used in real production environments.

Through AnalystWorld.in, Nikhil focuses on helping learners build practical analytical skills and prepare confidently for real data roles. His writing and mentorship aim to move students beyond tool tutorials into the reasoning that actual interviews test.

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If this book helped you even a bit, do let me know...



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